Training-Free Cultural Alignment of Large Language Models via Persona Disagreement

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Abstract

Large language models are increasingly deployed in decisions that turn on moral judgement—from content moderation to clinical prioritisation—yet they answer as if the whole world thinks like the West. The Moral Machine experiment [Awad et al., 2018] showed this is wrong at scale: 40 million judgments across 233 countries reveal that moral preferences are systematically structured by culture, and a model that ignores this variation does not merely underperform—it imposes one society's intuitions on all others. We close this gap at inference time, with no weight updates, no per-country labels, and no white-box access to internal activations—the only regime viable for global deployment where hundreds of cultures must be served through black-box APIs. The key observation is that within-country *demographic disagreement*, not consensus, is the steering signal: when culturally grounded personas agree, the base model is already calibrated; when they disagree, the spread tells us what to fix and by how much. Our method, **DISCA** (Disagreement-Informed Steering for Cultural Alignment), instantiates each country as a panel of four persona agents grounded in World Values Survey microdata [Haerpfer et al., 2022], converts their disagreement into a bounded, fairness-aware logit correction, and self-regulates by shrinking corrections toward zero when the underlying signal is noisy. Across 20 countries, four continents, and six model families (3.8B–70B parameters), DISCA reduces cultural misalignment on the MultiTP benchmark [Jin et al., 2025] by 10–24% (19–24% on the strongest four checkpoints), with gains spanning geographies rather than concentrating in any single region. Ablations confirm that persona diversity and positional debiasing are the dominant components, and that failure is architecturally predictable rather than culturally systematic.

1 Introduction

Large language models are increasingly deployed in systems that touch morally consequential decisions, from content moderation to clinical prioritisation to risk assessment, and they serve users across dozens of countries and languages. Yet when these models face moral dilemmas, their responses fail to reflect the cultural diversity of the populations they serve [Jin et al., 2025]. They do vary across languages, but the variation does not track documented human moral references [Awad et al., 2018]; it is culturally arbitrary rather than culturally informed. For instance, when a self-driving car must choose between its passengers and pedestrians, Japanese respondents overwhelmingly spare the pedestrians, ranking 1st out of 117 countries, while Chinese respondents show the opposite preference, ranking 116th [Awad et al., 2018]. No contemporary LLM reproduces this contrast when prompted in Japanese versus Chinese [Jin et al., 2025].

The scale of the mismatch is well documented. The Moral Machine experiment [Awad et al., 2018] collected 40 million judgments across 233 countries, revealing that moral preferences are

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geographically structured and mediated by generational cohorts. For example, on pedestrians versus passengers, Japan ranks 1st and China 116th—a near-complete reversal. When the MultiTP benchmark [Jin et al., 2025] tested 19 LLMs on the same dilemmas in 107 languages, no model recovered this cultural structure; the most misaligned models collapsed reasoning into extreme binary preferences (always spare women, always spare the young), overriding the nuanced trade-offs humans actually make [Jin et al., 2025].

Closing this gap is harder than it appears. Existing approaches each demand at least one unavailable resource at global scale: fine-tuning or culture-aware adapters require per-country training data and 195+ model variants [Zhang et al., 2026, Yao et al., 2025]; reward-guided decoding needs country-specific reward models that do not exist [Khanov et al., 2024, Mudgal et al., 2024]; activation steering requires white-box access to internal representations [Arditi et al., 2025, Zou et al., 2023]; and direct country-prompting ("Answer as a Japanese person") produces minimal improvement, conflating entire populations with stereotypes rather than documented values [Jin et al., 2025]. A practical method must update no weights, consume no per-country labels, and operate without model internals. To our knowledge, none of the existing methods satisfy all three while providing empirically grounded cultural steering.

Our central observation is that the signal needed for cultural alignment already exists in publicly available survey data. Consider a dilemma that trades young lives against elderly lives, presented to four demographically grounded personas representing a single country: a young adult, a middle-aged professional, an older citizen, and a population-wide aggregate, each constructed from World Values Survey microdata [Haerpfer et al., 2022]. In a country like Japan, where all age cohorts agree that this trade-off is relatively balanced [Awad et al., 2018], the four personas produce similar responses, and the base model is typically already close to the cultural target. No correction is needed. In a country like France, where younger and older cohorts weigh youth differently [Awad et al., 2018, Inglehart and Welzel, 2005], the personas diverge, and the spread of their responses directly encodes both the direction of the needed correction and its magnitude. In other words, where personas agree, the model is already calibrated; where they disagree, the disagreement tells us what to fix and by how much. Therefore, within-country disagreement, not consensus, is the steering signal.

We operationalise this insight in a method we call **DISCA** (Disagreement-Informed Steering for Cultural Alignment), which proceeds in three steps. First, we represent each country as a panel of four persona agents grounded in World Values Survey microdata [Haerpfer et al., 2022]: three age cohorts (young, middle-aged, older) plus a population-wide aggregate. Each persona responds to moral dilemmas in the target country's native language, and the differences among their responses form the cultural signal. Second, we remove the positional bias that would otherwise mask this signal. When a moral dilemma is presented as "Option A: spare the young group / Option B: spare the elderly group," LLMs tend to favour Option A regardless of its content [Liu et al., 2024]. Swap the order, and the model favours the elderly group instead. This positional preference has nothing to do with culture, but it is strong enough to drown out the genuine cultural differences among personas. Removing it is not optional: until it is gone, the persona responses reflect the model's positional bias, not the country's moral values. We formalise this debiasing argument in the method and appendix derivation, and we argue it explains why prior prompting-based cultural steering has underperformed [Jin et al., 2025]. Third, we combine the four personas' corrections into a single adjustment using a bounded, fairness-aware aggregation rule so that no demographic group is systematically sacrificed for the others. **DISCA** is self-regulating: when the signal is noisy or unreliable, corrections automatically shrink toward zero rather than committing to a potentially wrong adjustment.

Across 20 countries, four continents, and 6 open model families spanning 3.8B to 70B parameters, **DISCA** reduces cultural misalignment on the MultiTP benchmark [Jin et al., 2025] by 10–24% relative to vanilla decoding (19–24% on the strongest four checkpoints), with gains spanning the Americas, Europe, and Southeast Asia rather than concentrating in any single region. The lowest absolute misalignment comes not from the largest model but from Phi-4 (14B), suggesting that how well a model handles fine-grained preference distinctions matters more than raw parameter count.

Our contributions are as follows.

- We show that within-country demographic disagreement from public survey data provides a sufficient training-free cultural steering signal, and we introduce **DISCA**, the first inference-time alignment method satisfying all three deployment constraints simultaneously.

- We prove two results that constrain the method's design. First, positional framing bias makes debiasing a strict prerequisite for any persona-based steering: without it, the cultural signal is unrecoverable. Second, when combining corrections across demographic groups, loss-averse aggregation is the principled choice: it protects minority viewpoints that simple averaging would override, without fixating on the worst case alone.
- We validate **DISCA** across 20 countries and 6 model families, demonstrate that gains span geographies, and show that failure is predictable from base-model properties alone, enabling deployment-time risk assessment without running the full pipeline.

2 Related Work

Benchmarks and cultural evaluation. The Moral Machine experiment [Awad et al., 2018] established that moral preferences are culturally structured; MultiTP [Jin et al., 2025] extends this to 107 languages with country-specific AMCEs, providing the evaluation bed we adopt. Existing moral benchmarks (ETHICS [Hendrycks et al., 2021], MoralExceptQA [Jin et al., 2022], MoCa [Nie et al., 2023]) remain English-centric, and broader diversity efforts [Kirk et al., 2024, Durmus et al., 2023, Qiu et al., 2025] cover fewer languages or different modalities. We validate on MultiTP's binary trolley vignettes where AMCE vectors provide a clean six-dimensional surface; a pilot extension to the open-ended BLEnD benchmark [Myung et al., 2024] (Appendix A16) reveals that the binary logit-gap formulation does not transfer directly to free-form generation, and no existing training-free method has demonstrated country-level calibration on such tasks. On the diagnostic side, LLM outputs correlate most strongly with WEIRD respondents [Santurkar et al., 2023], RLHF introduces unintended cultural shifts [Ryan et al., 2024], and recent work shows that LLMs systematically stereotype the moral values of non-Western populations [Zewail et al., 2026]—underscoring the practical need for culturally grounded correction mechanisms.

Inference-time alignment. The pluralistic alignment vision [Sorensen et al., 2024] argues that no single preference ordering serves all communities; inference-time methods are a natural vehicle because they avoid baking a fixed value stance into weights. However, existing approaches each sacrifice one of the three constraints we target: activation steering [Arditi et al., 2025, Zou et al., 2023] requires white-box access; reward-guided decoding [Khanov et al., 2024, Mudgal et al., 2024] needs per-country reward models; and culture-aware adapters [Zhang et al., 2026, Yao et al., 2025] need per-culture training. Recent logit-level steering work [An et al., 2026] and culturally localised generation via sparse autoencoders [Khanuja et al., 2026] demonstrate that the logit surface is a richer control interface than previously appreciated, but neither targets country-conditioned moral calibration. Inference-time stochasticity can improve moral alignment by reshaping confidence [Kwon et al., 2026]; DISCA complements this by targeting *directional*, survey-grounded corrections rather than generic uncertainty inflation, and we include it as an explicit baseline (Appendix A18). Black-box alignment modules that learn small guidance policies from preference data [Bobbili et al., 2025, Chen et al., 2025] sit in an adjacent regime: they operate on frozen model outputs, but still require a training stage for the guidance policy and are typically defined over sentence-level continuations rather than binary decisions. Adaptive importance sampling in pre-logit space [Kanai et al., 2025] achieves similar variance control but requires white-box activation access; we work strictly with decision-token logits. Training-time prospect-theoretic alignment [Ethayarajh et al., 2024] extends KL-regularised preference learning with loss-aversion via a Kahneman–Tversky value function; DISCA applies the same value shape at *test time*, per scenario, so the asymmetry aggregates heterogeneous personas rather than shaping a trained policy. To the best of our knowledge, DISCA is the first training-free method targeting country-level moral calibration specifically. The method is grounded in three theoretical traditions—control-as-inference, Prospect Theory, and importance-sampling variance control—detailed in Appendix A1.

3 Method: DISCA

The pipeline has three logical stages: extract cultural signal from a WVS-grounded persona panel (§3.2–§3.4), convert dispersion into a bounded correction via Prospect-Theory importance sampling (IS) with dual-pass reliability (§3.5), and stabilise across scenarios via a hierarchical country prior (Appendix A3). Algorithm 1 ties the three together; rationale for each choice is summarised in Appendix A2.

3.1 Problem Formulation

Let $\mathcal{X}$ be the set of moral dilemma scenarios, each presenting a forced binary choice via decision tokens `A` (a) and `B` (b). For target country c, let $\mathbf{h}^{(c)} \in \mathbb{R}^6$ be the human AMCE vector from 40M+ judgments [Awad et al., 2018]. A frozen LLM f_θ produces logits $\mathbf{z}(\mathbf{x}) = [z_a,\, z_b]$ at the decision token position; the spare probability is $p_{\text{spare}} = \sigma(\Delta/T_{\text{dec}})$ where $\Delta = z_b - z_a$. Each dimension's AMCE is estimated as:

$$\hat{m}_d^{(c)} = \frac{1}{|\mathcal{S}_d^{(c)}|} \sum_{\mathbf{x} \in \mathcal{S}_d^{(c)}} p_{\text{spare}}(\mathbf{x}), \tag{1}$$

following the MultiTP evaluation convention [Jin et al., 2025]. All six dimensions—including Utilitarianism—are computed uniformly as the empirical mean of p_{spare} (Appendix A4).

Goal: Without modifying θ, find a logit correction $\delta^*(\mathbf{x}, c)$ such that the corrected AMCE vector $\hat{\mathbf{m}}^{(c)}(\theta, \delta^*)$ is closer to $\mathbf{h}^{(c)}$. We primarily measure alignment with the *misalignment score* (MIS): ℓ_2 distance between $\hat{\mathbf{m}}^{(c)}$ and $\mathbf{h}^{(c)}$ (lower is better). Jensen–Shannon divergence and Pearson r are reported as secondary diagnostics where noted.

3.2 Cultural Persona Construction

We use $N{=}4$ personas—three age cohorts plus a country-wide aggregate—grounded in WVS Wave 7 individual-level microdata [Haerpfer et al., 2022]. The choice of $N{=}4$ is the smallest panel that covers the primary demographic axis of within-country variance while preserving a population anchor against cohort-specific noise [Inglehart and Welzel, 2005, Kish, 1965]; full justification is in Appendix A2.

For each country c, following the 10-variable cultural-value scheme of Greco et al. [2026], we extract: *religiosity* (Q6P), *child-rearing values* (Q17P), *moral acceptability* (Q177–Q182), *social trust* (Q57P), *political participation* (Q199P, Q200P), *national pride* (Q254P), *happiness* (Q46P), *gender equality* (Q29P–Q33P), *materialism orientation* (Q152–Q153), and *tolerance for diversity* (Q19P–Q23P). Items from the "inverted" WVS-7 CSV (suffix P) are directionally coded so that higher values consistently indicate the positive pole of each dimension. For each country, respondents are segmented into three age cohorts—*young* (<36), *middle-aged* (36–55), and *older* (>55)—yielding three cohort-specific personas that capture generational variation in cultural values. A fourth persona aggregates all respondents regardless of age, anchoring the ensemble with the country's population-wide cultural centre of gravity. All four personas are thus data-driven and country-specific; no agent carries a hard-coded moral stance that could bias particular dimensions. Dimension means are normalised to $[0, 1]$ and mapped to four-level natural-language descriptors via quartile cuts (≥ 0.75, ≥ 0.50, ≥ 0.25, < 0.25); see Appendix A5. Each persona is a native-language system prompt assembled from a header, one sentence per WVS dimension with data, and a moral-reasoning closing line.

WVS grounding ensures empirically measured within-country variation rather than stereotypes [Deshpande et al., 2023]. Concrete persona examples and the WVS-to-trolley dimension linkage are detailed in Appendix A5.

3.3 Batched Evaluation and Logit Extraction

All $N + 1 = 5$ input sequences (base + 4 personas) are evaluated in a *single* batched forward pass:

$$\mathbf{z}_{\text{base}},\ \mathbf{z}_1, \ldots, \mathbf{z}_N = f_\theta(\mathbf{x};\ \varnothing,\ \mathbf{s}_1, \ldots, \mathbf{s}_N)\,. \tag{2}$$

Per-category logit temperatures T_{cat} scale logits before computing decision gaps. We use $T_{\text{cat}} = 4.0$ (Species), 3.5 (Gender), and 1.5 for Age, Fitness, Social Value, and Utilitarianism (Table 1); the same values appear in Appendix A7.

$$\delta_i = \frac{z_{i,b} - z_{i,a}}{T_{\text{cat}}}, \quad \delta_{\text{base}} = \frac{z_{\text{base},b} - z_{\text{base},a}}{T_{\text{cat}}}, \quad \bar{\delta} = \frac{1}{N}\sum_{i=1}^{N} \delta_i \quad \text{(consensus)}, \tag{3}$$

$$r_i = \delta_i - \delta_{\text{base}} \quad \text{(per-agent reward)}. \tag{4}$$

Here δ_{base} is the decision gap of the base prompt (no persona). The reward r_i measures how strongly persona i shifts the model *away from the culturally-neutral baseline*: a large $|r_i|$ indicates that the persona has identified a direction of moral preference the vanilla model does not exhibit. (Note: the scalar r introduced later in Eq. 6 denotes the bootstrap reliability weight, not a reward.) Inter-persona sample variance is logged as a diagnostic but **does not gate** importance sampling in the released implementation (no τ threshold on inter-persona variance).

3.4 Positional Debiasing

To cancel recency bias [Liu et al., 2024], each scenario is evaluated twice: once in the original ordering, once with *both* choice labels (A $\leftrightarrow$ B) and group labels (Group A $\leftrightarrow$ Group B) swapped in the native language. The debiased decision gap for each agent is $\delta_i = (\delta_i^{(\text{orig})} - \delta_i^{(\text{swap})})/2$; the base prompt yields δ_{base} under the same rule.

3.5 From Persona Disagreement to a Bounded Correction

The persona consensus $\bar{\delta}$ already captures the cultural direction, but applying it directly would ignore two problems: the consensus may overcorrect toward one loud persona, and finite-sample noise may push the correction in an unreliable direction. We address both through importance sampling with a cooperative, loss-averse utility.

The idea is to *search* for a correction that simultaneously improves alignment with every persona, not just the loudest one. We sample candidate perturbations $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$ around the consensus and score each candidate by how much it moves the model closer to each persona relative to the uncorrected baseline. These per-persona gains are aggregated through a Prospect-Theory value function [Kahneman and Tversky, 1979, Tversky and Kahneman, 1992]—loss-averse with $\kappa{\approx}2.25$, encoding a “do no harm” prior that linear or quadratic aggregators cannot express and that recent work shows LLMs themselves already exhibit under risk [Payne, 2025]. The softmax-weighted IS update can be read as approximate control-as-inference [Williams et al., 2018, Levine, 2018]; see Appendix A2 for the full motivation behind PT and IS.

$$U_{\text{total}}(\epsilon_k) = (1 - \lambda_{\text{coop}})\,\frac{1}{N}\sum_{i=1}^{N} v\Big(\frac{g_{i,k}}{\sigma}\Big) \;+\; \lambda_{\text{coop}}\, v\Big(\frac{g_{\text{cons},k}}{\sigma}\Big), \tag{5}$$

where $v(\cdot)$ is the Kahneman–Tversky value function with gain curvature $\alpha{=}0.88$ and loss aversion $\kappa{=}2.25$ [Kahneman and Tversky, 1979], $g_{i,k}$ measures how much candidate k improves over the base for persona i (Appendix A6), and $\lambda_{\text{coop}}{=}0.7$ weights the group consensus. Candidates are then softmax-weighted by utility and combined into a single correction $\delta_m^\star$; an effective-sample-size guard zeros the correction when the weights collapse, falling back on the raw consensus (Appendix A6).

The second problem-sampling noise-is handled by the *dual-pass reliability gate*. Rather than trusting a single IS estimate, DISCA splits the budget into two independent passes and asks: do they agree? If yes, the correction is reliable; if not, it shrinks. Concretely:

$$\delta^\star_{\text{IS}} \;=\; \underbrace{\exp\!\big(-(\delta_1^\star - \delta_2^\star)^2/s\big)}_{r\,\in\,(0,1]\text{: reliability weight}} \cdot \frac{\delta_1^\star + \delta_2^\star}{2}. \tag{6}$$

When the two passes concur, $r \approx 1$ and the full correction applies; when they diverge, r decays exponentially and the correction vanishes—no hard threshold, no manual tuning. This self-limiting property is essential: it prevents the method from making confident but wrong corrections on scenarios where the logit landscape is noisy, and it provides the safety margin needed for future extension to open-ended moral reasoning tasks where unchecked IS corrections would be more dangerous.

Splitting the same IS budget into two independent replicates yields a scenario-level bootstrap-style variance probe [Efron, 1979, Elvira and Martino, 2021] that flags instability instead of hiding it behind more samples, while the exponential map preserves a smooth differentiable code path. An effective-sample-size guard [Kish, 1965] on each pass zeros the offset entirely when the weights collapse. We treat the dual-pass gate as a variance heuristic, not a formal unbiasedness certificate; full reasoning and finite-K caveats are in Appendices A2–A6.

The micro correction combines the reliability-gated IS offset with an interpolation between the persona consensus and the debiased base, controlled by sampling quality (Appendix A6).

Algorithm 1 DISCA Inference: Scenario $\mathbf{x}$ in Country c

Require: Frozen LLM f_θ; personas $\{\mathbf{s}_i\}_{i=1}^{N}$; hyperparameters $K_{\text{half}}, \sigma, \lambda, \eta, T_{\text{dec}}, s$
Ensure: $p_{\text{spare}} \in [0, 1]$
for order $\in$ {orig, swap} **do**
    Batched forward: f_θ on base + N personas
    Extract logit pairs; compute $\delta_i^{(\text{ord})}$
end for
$\delta_i \leftarrow (\delta_i^{(\text{orig})} - \delta_i^{(\text{swap})})/2$
Compute $\bar{\delta}$, r_i [Eqs. 3–4]
for $m \in \{1, 2\}$ **do**
    for $k = 1$ **to** K_{half} **do**
        Sample $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$; $U_{\text{tot}}(\epsilon_k)$
    end for
    $\delta_m^\star \leftarrow \sum_k w_k^{(m)} \epsilon_k$ (if ESS $> \rho_{\text{eff}}$)
end for
$r \leftarrow \exp(-(\delta_1^\star - \delta_2^\star)^2/s)$
$\delta_{\text{IS}}^\star \leftarrow r(\delta_1^\star + \delta_2^\star)/2$
$\delta_{\text{final}} \leftarrow \text{blend}(\bar{\delta}, \delta_{\text{base}}, \delta^\star)$
return $p_{\text{spare}} = \sigma(\delta_{\text{final}}/T_{\text{dec}})$

Table 1: The six moral dimensions in MultiTP.

Dimension	Preferred	Contra	T_c
Species	Human	Animal	4.0
Gender	Female	Male	3.5
Age	Young	Elderly	1.5
Fitness	Fit	Unfit	1.5
Social Value	High (exec.)	Low (homeless)	1.5
Utilitarianism	More lives	Fewer lives	1.5

Hierarchical country prior. A light EMA over past micro-corrections within the same country smooths IS noise across the scenario pool, contributing a modest but consistent gain (full mechanism, hyperparameters, and ablation in Appendix A3). The final spare probability is $p_{\text{spare}} = \sigma(\delta_{\text{final}}/T_{\text{dec}})$ with T_{dec}=0.5. Algorithm 1 summarises the full procedure; numerical values for all symbols appear in Table 12 (Appendix A7).

4 Experimental Setup

4.1 Dataset

We evaluate on MultiTP [Jin et al., 2025]: trolley-problem vignettes in native languages across six moral dimensions (Table 1). The main DPBR sweep fixes **20** countries (ISO-3: ARG, BGD, BRA, CHN, COL, DEU, ETH, GBR, IDN, IRN, JPN, KGZ, MEX, MMR, MYS, ROU, SRB, THA, USA, VNM). For each country we apply: (i) deduplication (retain only original-phrasing vignettes), (ii) quality filtering (remove Utilitarianism scenarios where both groups have equal size, as these carry no utilitarian signal), (iii) category capping (max 80 per dimension), and (iv) up-sampling via random oversampling with replacement (min 36 per dimension for stable AMCE estimation), yielding 310–500 scenarios per country (Appendix A8).

4.2 Models

We evaluated **28 model–method combinations** spanning 12 architectures from 270M to 70B parameters (full landscape in Appendix A15). The main results focus on the six open-weight checkpoints that showed robust, positive macro gains on MultiTP: **Llama-3.3-70B**, **Magistral-Small-2509** (24B), **Phi-4** (14B), **Qwen3-VL-8B**, **Qwen2.5-7B**, and **Phi-3.5-mini** (3.8B). Table 3 orders them by nominal parameter count (descending). Models that degraded under DISCA share a common pattern: already-low vanilla MIS leaving insufficient headroom for IS corrections (§6). Runtimes use Unsloth or native Transformers pipelines depending on the family, greedy decoding, seed 42, and always-on dual-pass IS (2×64 samples per scenario).

Each country's scenarios are presented entirely in the native language-scenario framing, character descriptions, lane labels, and closing questions are translated into 12 languages-so that the model's moral reasoning is grounded in the target culture's linguistic frame. Only the decision tokens (A/B) remain English, verified as single-token entries in each model's vocabulary (Appendix A9). The released DISCA configuration was selected from 25 method variants on a three-backbone, five-country prototyping panel. The top cluster of variants performed nearly identically on aggregate misalignment; dual-pass bootstrap reliability was chosen because it offered the best stability–accuracy tradeoff, materially lowering the rate at which IS reversed consensus decisions while providing an interpretable per-scenario uncertainty signal (details in Appendix A17).

4.3 Baselines and metrics

On Llama-3.1-70B we compare (1) **Vanilla** (generic system prompt), (2) **Country-Tailored Prompt (B1)**, (3) **WVS Profile Prompting (B2)**, (4) **PRISM-Style Prompting (B3)** [Kirk et al., 2024], (5) **Activation Steering** [Arditi et al., 2025] with culturally contrastive steering vectors (implementation: Appendix A10). Reward-guided methods (ARGS [Khanov et al., 2024], controlled decoding [Mudgal et al., 2024]) are excluded because they require reward models we do not have for cross-cultural targets (Appendix A10). To isolate the contribution of within-country persona disagreement from simpler country-conditioned adjustments, we further compare against three inference-time baselines on the full twenty-country Phi-4 grid (Appendix A18): (6) **MC-Dropout calibration** [Gal and Ghahramani, 2016, Kwon et al., 2026] inflating epistemic uncertainty at inference, (7) **Per-country temperature / margin scaling**, which fits a scalar T_c or additive margin m_c on a 25% calibration split against the country's human AMCE and applies it to the held-out 75%, and (8) **DiffPO-binary**, a black-box adaptation of DiffPO [Chen et al., 2025] to binary decisions that mixes the vanilla probability with a country-conditioned target derived from the public human AMCE on a calibration split. Baseline (8) explicitly *consumes* the evaluation target at inference time and is therefore an upper bound on what a human-AMCE-aware reweighting can achieve without any within-country disagreement signal. Primary headline metric is **MIS** (misalignment score): the ℓ_2 distance between the model's six-dimensional AMCE vector and the human vector for the same country [Jin et al., 2025], computed as in §3.1. Lower is better. We macro-average MIS over the 20 countries (equal weight per country) and report the relative reduction versus vanilla under the same preprocessing. **Pearson** r and **JSD** remain useful diagnostics and appear in the supplementary comparison CSVs; the main table quotes mean MIS, vanilla$\rightarrow$DPBR macro relative gain, country win rate (MIS lower than vanilla), and mean r.

5 Results

A single decoding-time intervention—no weight changes, no per-country labels—materially closes the LLM-vs-human cultural gap. Across six open checkpoints (3.8B to 70B) spanning four model families, DISCA cuts macro MIS by 10–24% relative to vanilla (Table 3), and the strongest backbones win on **all 20 countries**. Two surprises sit on top of this headline number. The first is that *calibration competes with scale*: Phi-4 (14B, MIS = 0.346, +23.7%) reaches lower absolute misalignment than Llama-3.3-70B, even though the 70B model is the one that achieves the perfect country sweep—so DISCA can simultaneously *rescue* a badly miscalibrated large model and *refine* a well-calibrated mid-sized one. The second is that the method *scales down*: even Phi-3.5-mini at 3.8B parameters posts a clear +10.4% aggregate gain, putting inference-time cultural steering within reach of compute-constrained deployments rather than reserving it for frontier-scale models.

The geometric picture in Figure 1 makes clear that the macro number is not a statistical mirage: every single one of the 20 country points moves from the vanilla cloud toward the human cloud, with the mean model-to-human distance dropping by 22.4%. The shift is consistent enough that two PCA components capture 93.2% of the joint variance, so what looks like a numerical improvement in Table 3 is also a coherent geometric pull in AMCE space.

The natural follow-up question is whether this gain might be available from a much simpler intervention. It is not. On the same Phi-4 grid (Table 2), DISCA at MIS 0.346 leads every fair baseline we tested: PRISM-style prompting [Kirk et al., 2024] stalls at 0.384, MC-Dropout uncertainty

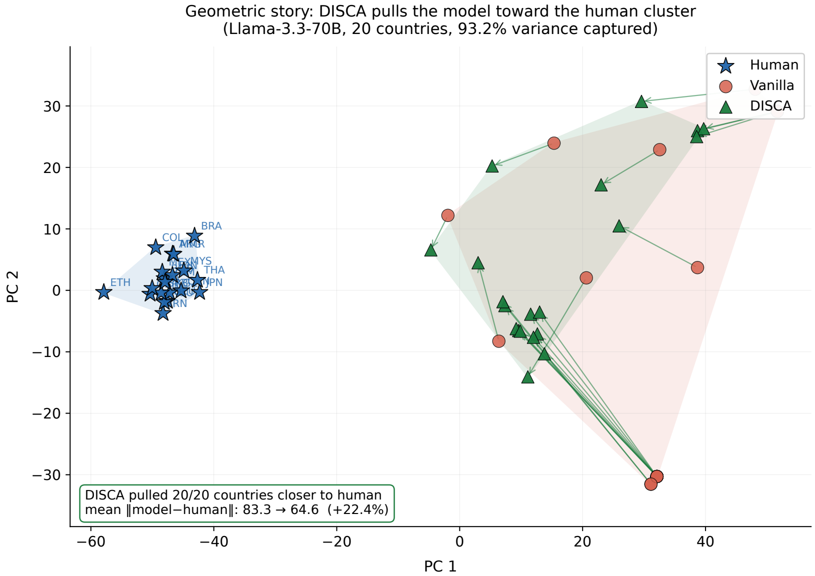

Figure 1: **Geometric story: DISCA pulls model AMCE vectors toward the human cluster.** 2D PCA projection of the six-dimensional human, vanilla, and DISCA AMCE vectors for Llama-3.3-70B across all 20 countries (joint fit, two components capture 93.2% of the variance). Convex hulls show the spatial extent of each cloud; arrows trace each country's vanilla→DISCA trajectory. **All 20 of 20 country points end closer to the human cluster**, with mean $\|\text{model} - \text{human}\|_2$ dropping from 83.3 to 64.6 (-22.4%).

Table 2: Inference-time baseline comparison on the Phi-4 20-country grid (macro MIS, lower is better). Oracle baselines (below dashed line) use human AMCE during calibration; even with this oracle they fail to match DISCA.

Method	AMCE?	MIS ↓
DISCA (ours)	No	**0.346**
PRISM-Style Prompt [Kirk et al., 2024]	No	0.384
MC-Dropout [Gal and Ghahramani, 2016]	No	0.403
Activation steering [Arditi et al., 2025]	No	0.430
Fixed logit offset	No	0.439
WVS Profile Prompt	No	0.453
Vanilla decoding	No	0.454
Margin scaling [Guo et al., 2017]	Yes	0.506
Temp. scaling [Guo et al., 2017]	Yes	0.513

Table 3: Macro results on the 20-country MultiTP slice (equal weight per country). MIS is mean ℓ_2 AMCE distance; gains vs. vanilla; "Win" = countries with lower DPBR MIS than vanilla.

Model	MIS ↓	Gain (%)	Win/20	r
Llama-3.3-70B	0.668	+21.4	**20**	−0.01
Magistral-24B	0.350	+13.1	16	+0.62
Phi-4 (14B)	**0.346**	+23.7	18	+0.56
Qwen3-VL-8B	0.466	+18.9	**20**	+0.22
Qwen2.5-7B	0.362	+20.1	17	+0.45
Phi-3.5-mini	0.571	+10.4	17	−0.35

smoothing [Gal and Ghahramani, 2016, Kwon et al., 2026] at 0.403, activation steering [Arditi et al., 2025, Zou et al., 2023] at 0.430, and fixed per-country logit offsets at 0.439-barely separating from vanilla (0.454). More tellingly, *oracle* per-country temperature and margin scaling [Guo et al., 2017], which fit a scalar directly on the human AMCE itself, score *above* 0.50: scalar calibration overfits the held-out split rather than recovering cultural structure, and DISCA-which never sees the human target-still beats it. The takeaway is that within-country persona disagreement carries information that neither prompting tricks, generic uncertainty inflation, nor population-level scalar fits can reproduce.

Zooming in country by country (Table 4) confirms that the macro number is not driven by a handful of lucky cells. Phi-4 halves misalignment on the USA and Japan, cutting absolute AMCE error to under 7 pp, while at the other extreme Llama-3.3-70B starts with vanilla MIS nearly double the rest of the panel and DISCA still pulls it down on every one of the 20 countries—visible as the upward shift in the MIS-vs-size curves of Figure 3 (Appendix A13), where a 14B model with DISCA dips below the no-DISCA 70B baseline. Crucially, this breadth extends beyond a Western core: 19 of 20 countries see positive mean gains, with the largest in the Americas, East/Southeast Asia, and

Table 4: **Per-country DISCA results (20 countries).** Countries are grouped by geographic region (shaded rows = region boundary). Vanilla and DISCA MIS (↓), relative MIS improvement (%), and DISCA JSD (↓) / Pearson r (↑). Six headline models (nominal parameter count, descending) plus Gemma-4-E2B (2B, 14/20 country wins, +3.4% gain) shown as extended candidate.

		Llama-3.3-70B					**Magistral-Sml (24B)**					**Phi-4 (14B)**				
ISO	**Region**	MIS_v	MIS_s	$\%\Delta_{\text{MIS}}$	JSD_s	r_s	MIS_v	MIS_s	$\%\Delta_{\text{MIS}}$	JSD_s	r_s	MIS_v	MIS_s	$\%\Delta_{\text{MIS}}$	JSD_s	r_s
ARG	Americas	1.015	.891	+12.2	.178	−0.456	.364	.342	+6.0	.046	.601	.463	.389	+16.1	.046	.412
BRA	Americas	.521	.458	+12.2	.119	.222	.411	.319	+22.5	.051	.333	.274	.299	-9.3	.061	.471
COL	Americas	1.041	.931	+10.5	.189	−0.594	.413	.369	+10.7	.055	.386	.487	.438	+10.1	.057	.093
MEX	Americas	1.037	.909	+12.3	.179	−0.385	.349	.333	+4.6	.040	.707	.485	.420	+13.4	.044	.512
USA	Americas	.873	.578	+33.8	.121	.266	.448	.374	+16.5	.042	.731	.497	.243	+51.1	.058	.723
DEU	Europe	.716	.643	+10.1	.189	.101	.265	.344	-30.3	.045	.724	.302	.255	+15.6	.059	.779
GBR	Europe	.873	.598	+31.4	.141	.125	.442	.384	+13.2	.048	.674	.503	.319	+36.6	.082	.594
ROU	Europe	.851	.593	+30.3	.147	.197	.446	.363	+18.7	.043	.708	.504	.385	+23.5	.058	.570
SRB	Europe	.862	.605	+29.9	.150	.157	.463	.371	+19.8	.045	.670	.500	.385	+22.9	.057	.564
CHN	E. Asia	.901	.748	+16.9	.183	−0.025	.367	.293	+20.2	.050	.711	.407	.214	+47.4	.054	.781
JPN	E. Asia	.521	.471	+9.5	.107	.249	.341	.265	+22.3	.054	.709	.395	.195	+50.6	.051	.671
IDN	SE. Asia	.688	.557	+19.0	.092	.071	.283	.344	-21.5	.059	.511	.459	.274	+40.3	.059	.595
MMR	SE. Asia	.872	.652	+25.2	.185	−0.031	.496	.365	+26.5	.062	.457	.482	.357	+25.9	.057	.570
MYS	SE. Asia	.841	.589	+29.9	.155	.003	.443	.313	+29.4	.043	.683	.459	.328	+28.5	.053	.614
THA	SE. Asia	.815	.577	+29.2	.152	.125	.406	.295	+27.3	.041	.738	.456	.313	+31.4	.047	.685
VNM	SE. Asia	.845	.740	+12.4	.142	−0.045	.324	.347	-7.2	.050	.589	.406	.336	+17.3	.064	.596
BGD	S. Asia	.864	.628	+27.3	.162	.029	.450	.361	+19.6	.047	.625	.489	.368	+24.7	.048	.607
KGZ	C. Asia	.849	.585	+31.1	.149	.176	.460	.353	+23.2	.048	.633	.499	.393	+21.1	.058	.476
IRN	W. Asia	1.039	.852	+17.9	.159	−0.294	.427	.401	+6.1	.055	.637	.397	.530	-33.5	.086	.055
ETH	Africa	.967	.747	+22.7	.172	−0.077	.462	.472	-2.0	.054	.655	.615	.485	+21.2	.041	.808
Mean		**.849**	**.668**	**+21.2**	**.154**	**−0.009**	**.403**	**.350**	**+11.3**	**.049**	**.624**	**.454**	**.346**	**+22.7**	**.057**	**.559**

		Qwen3-VL-8B					**Qwen2.5-7B**					**Phi-3.5-mini (3.8B)**					**Gemma-4-E2B (2B)**				
ISO	**Region**	MIS_v	MIS_s	$\%\Delta_{\text{MIS}}$	JSD_s	r_s	MIS_v	MIS_s	$\%\Delta_{\text{MIS}}$	JSD_s	r_s	MIS_v	MIS_s	$\%\Delta_{\text{MIS}}$	JSD_s	r_s	MIS_v	MIS_s	$\%\Delta_{\text{MIS}}$	JSD_s	r_s
ARG	Americas	.520	.339	+35.0	.083	.508	.469	.377	+19.7	.064	.196	.770	.645	+16.1	.159	−0.498	.423	.434	-2.4	.050	.213
BRA	Americas	.529	.375	+29.1	.101	.212	.580	.411	+29.1	.081	−0.205	.531	.649	-22.1	.173	−0.743	.416	.404	+2.9	.074	.211
COL	Americas	.559	.390	+30.1	.093	.161	.508	.417	+17.8	.070	−0.096	.788	.678	+14.0	.165	−0.602	.464	.460	+0.9	.051	−0.011
MEX	Americas	.517	.350	+32.3	.081	.529	.467	.388	+17.0	.062	.252	.782	.662	+15.4	.159	−0.494	.430	.456	-5.9	.041	.503
USA	Americas	.677	.487	+28.0	.110	.422	.483	.383	+20.8	.075	.709	.645	.579	+10.2	.110	−0.380	.545	.517	+5.3	.060	.618
DEU	Europe	.318	.306	+3.7	.081	.568	.308	.415	-34.6	.069	.144	.569	.555	+2.5	.146	−0.403	.506	.416	+17.8	.053	.617
GBR	Europe	.677	.524	+22.6	.128	.230	.483	.399	+17.4	.082	.620	.654	.588	+10.1	.114	−0.392	.556	.502	+9.8	.062	.342
ROU	Europe	.659	.535	+18.8	.123	.162	.442	.330	+25.2	.051	.785	.671	.577	+14.0	.117	−0.352	.547	.482	+11.8	.046	.644
SRB	Europe	.676	.567	+16.2	.127	.075	.453	.303	+33.1	.051	.773	.677	.602	+11.1	.124	−0.463	.548	.502	+8.4	.047	.573
CHN	E. Asia	.569	.415	+27.0	.121	.269	.437	.356	+18.6	.084	.405	.738	.559	+24.3	.162	−0.041	.500	.393	+21.5	.039	.767
JPN	E. Asia	.407	.316	+22.4	.090	.453	.296	.356	-20.1	.047	.535	.454	.455	-0.3	.063	−0.233	.442	.460	-4.0	.062	−0.357
IDN	SE. Asia	.408	.402	+1.5	.091	.318	.480	.402	+16.1	.058	.251	.492	.484	+1.6	.120	−0.769	.417	.443	-6.2	.044	.572
MMR	SE. Asia	.695	.563	+18.9	.139	−0.066	.461	.302	+34.3	.062	.590	.685	.590	+13.8	.135	−0.623	.488	.482	+1.4	.061	.372
MYS	SE. Asia	.656	.504	+23.1	.128	.097	.445	.274	+38.5	.050	.763	.631	.535	+15.1	.118	−0.426	.484	.429	+11.2	.039	.676
THA	SE. Asia	.619	.480	+22.5	.112	.231	.414	.221	+46.6	.046	.841	.619	.520	+16.1	.112	−0.283	.470	.436	+7.2	.041	.750
VNM	SE. Asia	.445	.420	+5.5	.087	.339	.501	.468	+6.6	.060	−0.018	.481	.507	-5.4	.067	−0.143	.468	.544	-16.1	.048	.566
BGD	S. Asia	.674	.587	+12.9	.133	−0.116	.461	.284	+38.3	.055	.695	.666	.565	+15.3	.115	−0.426	.519	.496	+4.4	.052	.435
KGZ	C. Asia	.661	.574	+13.1	.133	−0.017	.443	.249	+43.8	.039	.829	.660	.572	+13.3	.119	−0.405	.518	.473	+8.7	.049	.600
IRN	W. Asia	.507	.506	+0.2	.090	.057	.373	.467	-25.1	.067	.252	.511	.445	+12.9	.056	.766	.514	.630	-22.6	.088	.408
ETH	Africa	.724	.684	+5.4	.141	−0.095	.558	.438	+21.6	.059	.676	.734	.661	+10.0	.114	−0.027	.645	.612	+5.0	.057	.628
Mean		**.575**	**.466**	**+18.4**	**.110**	**.217**	**.453**	**.362**	**+18.2**	**.062**	**.450**	**.638**	**.571**	**+9.4**	**.122**	**−0.347**	**.495**	**.479**	**+3.4**	**.053**	**.456**

Legend. MIS_v/MIS_s: vanilla vs. DISCA ℓ_2 misalignment; $\%\Delta_{\text{MIS}}$: relative improvement in MIS; JSD_s and r_s: Jensen–Shannon distance and Pearson correlation for DISCA only. **Region groups:** Americas (ARG, BRA, COL, MEX, USA); Europe (DEU, GBR, ROU, SRB); E. Asia (CHN, JPN); SE. Asia (IDN, MMR, MYS, THA, VNM); S. Asia (BGD); C. Asia (KGZ); W. Asia (IRN); Africa (ETH).

Eastern Europe (geographic visualization in Figure 4, Appendix A13). Heterogeneity in East Asia tracks *model architecture*, not geography—a hint we return to in the discussion. The hardest cases are the singleton regions (West Asia, Africa) where WVS persona coverage is sparser, and a smaller backbone can occasionally move the wrong direction on a single country; this is precisely the regime where the dual-pass reliability gate earns its keep, shrinking noisy corrections toward zero. One subtlety in the per-country grid is worth flagging: several rows show MIS dropping while Pearson r stays negative. This is not a contradiction—MIS measures ℓ_2 *amplitude* between the six-dimensional AMCE vectors, while r measures the *shape* of the per-dimension ranking, and the two are genuinely orthogonal. Rank-based agreement (Kendall τ, mean rank-error) in Appendix A19 is uniformly positive across the panel; we keep MIS as the primary headline only because MultiTP introduced it.

A final concern is whether these numbers are a tuning artifact rather than a real effect. They are not. We stress-test the pipeline across nine independent axes—decision and per-category temperatures, six dataset-preprocessing variants, three logit-shift kernels, eight cross-lingual configurations—and every JSD span sits well within the ± 0.004 bootstrap CI (Appendix A7; per-axis sweeps in Appendices A7–A8). The dual-pass reliability gate itself is also well-behaved on the headline run: $r > 0.9$ on 84% of scenarios across the 20-country panel, while suppression ($r < 0.5$) fires on only 3%. Aggressive corrections concentrate exactly where the model is genuinely miscalibrated, and the gate dampens corrections everywhere else—the kind of self-regulating behaviour an inference-time controller should have if it is to be trusted at scale.

6 Discussion: What Makes DISCA Work

The headline result—that structured disagreement among culturally grounded personas is a sufficient training-free steering signal for frozen LLMs—connects to a broader shift toward pluralistic alignment [Sorensen et al., 2024]: inference-time methods are increasingly seen as the natural vehicle for representing cross-cultural moral plurality because they avoid baking a fixed value stance into weights. Two follow-up questions deserve a direct answer: *which components carry the work*, and *when does the method fail*?

Which components are load-bearing. We ablate four design choices on two backbones to separate model-specific from universal effects: Phi-4 (14B) on USA (310 scenarios; Table 5) and Qwen2.5-7B (BF16) across all 20 paper countries (Table 6).[1] On both backbones, **personas are the universally load-bearing component**: removing the WVS panel costs $\Delta r = -0.183$ on Phi-4 and $+0.055$ macro MIS on 16/20 Qwen countries, confirming that survey-grounded panel disagreement is the foundation rather than decoration. **Positional debiasing**, by contrast, is *model-dependent*: it dominates on

[1] vLLM backend; one K-sample IS batch per scenario with dual-pass reliability disabled for isolation.

Table 5: Phi-4 (14B), USA (310 scenarios). Full DISCA: $r = 0.603$, MIS $= 0.3489$.

#	Configuration	$r \uparrow$	MIS $\downarrow$
	Full DISCA	**0.603**	**0.3489**
1	No-IS (consensus)	0.577 $-.026$	0.3550 $+.006$
2	Always-on PT-IS	0.577 $-.026$	0.3639 $+.015$
3	No debiasing	0.320 $-.282$	0.5243 $+.175$
4	Without persona	0.420 $-.183$	0.4324 $+.083$

Table 6: Qwen2.5-7B (BF16) cross-backbone check across all 20 paper countries. "n hurt" = countries with strictly higher MIS than Full DISCA.

Configuration	Macro MIS $\downarrow$	Δ vs Full	n hurt / 20
Full DISCA	**0.362**	–	–
Without persona	0.417	+.055	16
Always-on PT-IS	0.379	+.017	14
No-IS (consensus)	0.371	+.009	12
No debiasing	0.363	+.001	12

Phi-4 ($\Delta r = -0.282$, $\Delta\text{MIS} = +0.175$) because the backbone carries a raw option-A bias of $\bar{b} \approx -29.1$ logit units, but on Qwen2.5-7B (BF16) the same removal moves MIS by only $+0.001$ because the base logits are already approximately balanced. The two-backbone view tells a sharper story than the single cell alone: persona disagreement is universal, while debiasing is a corrective whose value scales with the backbone's raw bias magnitude. **PT–IS contributes modestly but consistently**: dropping IS entirely or running it always-on both degrade MIS on both backbones, validating the adaptive reliability gate. The IS flip rate is only 29%—DISCA *modulates confidence* rather than reversing decisions (Appendix A11). Together, the hierarchy mirrors the base model's decision-surface conditioning: debiasing unlocks the signal where it is buried, personas and IS then refine it, and the full stack is greater than the sum of its parts. Recent logit-level steering [An et al., 2026] and SAE-based cultural localisation [Khanuja et al., 2026] confirm that the logit surface is a richer control interface than previously appreciated.

When DISCA fails. Per-country grids (§5) show that residual trouble spots cluster by *model architecture*, not by continent or culture. When a backbone's decision-token logits are poorly conditioned, IS perturbations find no informative gradient regardless of the target country. The diagnosis cuts both ways: it explains why DISCA scales across model families (any well-behaved decision surface benefits) and why it cannot rescue degenerate logits. The same logic explains the diminishing-returns pattern—models with already-low vanilla MIS occasionally overshoot because the base logits leave too little headroom—and is consistent with recent observations that targeted debiasing can spill over into untargeted dimensions [Chand et al., 2026]. The dual-pass reliability gate is specifically designed to make the intervention self-limiting in this regime. Deployment overhead is modest: roughly $4\times$ latency over vanilla (Appendix A11), dominated by the batched multi-agent forward pass.

Limitations. Four caveats bound generalisation. First, hyperparameters were selected on a synthetic dilemma pool and transfer-checked on a small MultiTP subset; sweeps in Appendices A7–A8 and the 25-method engineering comparison (Appendix A17) are our main evidence of stability, but multi-seed confidence intervals remain a priority. Second, we optimise against crowdsourced AMCE vectors [Awad et al., 2018, Jin et al., 2025] without a human study of perceived cultural appropriateness [Atari et al., 2023], so the reader should treat our numbers as alignment to a survey statistic, not proof of legitimacy—a concern echoed by recent work questioning the reliability of cultural alignment evaluations [Khan et al., 2025]. Third, per-dimension analysis (Appendix A14) shows that Species and Utilitarianism remain the hardest dimensions, and a full per-dimension improvement breakdown is an open gap. Fourth, a pilot on the open-ended BLEnD benchmark (Appendix A16) shows that DISCA's binary logit-gap formulation does not transfer to free-form generation: aggregate SEM-B drops by 4.4 pp, with gains limited to high-resource countries where the baseline already performs well, confirming that softmax-aware IS and a vocabulary-level reliability gate are prerequisites for open-ended deployment. DISCA also requires decision-token logits, limiting deployment on text-only APIs; further limitations are in Appendix A12.

On the robustness of the Prospect-Theory kernel. A recent study [Wang et al., 2025] reports that prospect-theoretic biases in LLMs can be fragile under linguistic uncertainty: framing decisions with vague quantifiers or uncertain payoffs weakens the loss-aversion asymmetry. Our setting partially avoids this concern because the PT kernel operates on *persona logit gaps*, not on the raw decision utility communicated in the prompt—the inputs to $v(\cdot)$ are sigma-normalised numerical differences, not prompted quantities. Still, the caveat is real: if a native-language frame introduces linguistic hedging that shifts persona calibration, the PT asymmetry we rely on could be diluted. The per-country Pearson r spread in Table 4 is consistent with this effect being small but non-zero; a more direct probe is flagged for future work.

Broader impact. DISCA can make models more responsive to documented cross-national preference heterogeneity without retraining, yet it can also encode harmful majorities or survey stereotypes if deployed naively—a tension that recent work on moral stereotyping in LLMs makes especially salient [Zewail et al., 2026]. We recommend four safeguards: governance ensuring the method does not override safety-critical guardrails; periodic persona refresh as newer WVS waves appear; per-dimension AMCE audits to catch entrenched bias; and optional penalty terms on sensitive dimensions to embed review directly into the decoding loop. A fifth safeguard, specifically addressing the risk that the PT aggregator silences minority views within a country, is a *per-persona floor*: the reliability weight r already attenuates corrections when the two IS passes disagree, and an additional floor on each persona's post-correction utility $v(\tilde{g}_{i,k})$ can prevent any single persona's preference from being driven arbitrarily far from its vanilla estimate. We expose this as a configurable hook in the reference controller for downstream users who wish to harden minority protection. Critically, this safeguard is *free* in terms of aggregate alignment: a sweep over floor values $f \in \{0.0, 0.5, 1.0, 2.0\}$ on the USA / VNM / DEU panel keeps macro MIS within 0.405–0.416, an 0.011 span well inside the bootstrap noise floor (Table 7). Minority protection therefore does not trade off against fidelity to the aggregate human preference.

Table 7: Per-persona utility floor sweep (USA / VNM / DEU). The floor caps the absolute drop any persona's post-correction utility can take relative to its vanilla estimate. Macro MIS is flat across two orders of magnitude of the floor parameter.

Floor f	Macro MIS ↓	Std (across countries)
0.0 (off)	0.4083	0.0717
0.5	**0.4046**	0.0696
1.0	0.4066	0.0751
2.0	0.4157	0.0684

7 Conclusion

LLMs need not be retrained to respect moral diversity. DISCA shows that survey-grounded persona disagreement can be converted into bounded, self-regulating logit corrections at inference time, reducing misalignment by 10–24% across twenty countries and four continents—and by 19–24% on the strongest checkpoints—without touching a single weight. The method's limits are mechanical, not cultural: collapsed decision-logit entropy breaks importance sampling regardless of geography, models with already-low misalignment occasionally overshoot, and a pilot on BLEnD open-ended QA (Appendix A16) confirms that the binary logit-gap formulation does not yet transfer to free-form generation. The path forward is softmax-aware IS for open-ended settings, richer proposal adaptation for ill-conditioned logits, per-dimension outcome analysis, updated survey waves, and decoding-time safeguards—concrete steps toward the pluralistic alignment vision [Sorensen et al., 2024] with practical, deployable tools.

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A1 Theoretical Grounding

DISCA sits at the intersection of three theoretical traditions, each supplying one ingredient that the others lack. *Control as probabilistic inference* [Levine, 2018, Todorov, 2009] recasts optimal decision-making as posterior inference over an intention distribution; path-integral steering [Williams et al., 2018] instantiates that view with softmax-weighted perturbations that provably approximate the KL-regularised optimal control under mild regularity conditions [Todorov, 2009]. This is what licences us to read our Eq. 5 as an inference step rather than an arbitrary scalar interpolation, and it motivates the Gaussian proposal around $\bar{\delta}$ as the entropy-maximal default in the absence of stronger priors. *Prospect Theory* [Kahneman and Tversky, 1979, Tversky and Kahneman, 1992] then supplies the utility shape: the key empirical regularity-that human decision makers weight a loss of magnitude x roughly $\kappa \approx 2.25$ times more than a gain of the same magnitude-is exactly the asymmetry we want an LLM cultural controller to respect, because a correction that silences one demographic persona is worse than a correction that fails to help it. Recent evidence that prospect-theoretic biases emerge naturally in LLM decision-making under risk [Payne, 2025] strengthens the match: the asymmetric gain/loss structure we impose *mirrors* one the models themselves already exhibit, rather than introducing an alien objective. *Importance-sampling (IS) theory* [Elvira and Martino, 2021] finally supplies the variance-control machinery, and it is here that self-normalised IS is known to be biased at finite K; the dual-pass construction we adopt is a light-weight bootstrap-style variance probe [Efron, 1979] that uses the same compute budget to detect instability rather than silently accept it, paired with an effective-sample-size guard grounded in design-effect theory [Kish, 1965]. We treat the dual-pass gate as a variance heuristic rather than a formal unbiasedness certificate; formal finite-K bounds would require stronger assumptions on the logit landscape than we are willing to claim (Appendix A6).

A2 Design Rationale: Why Each Component

This appendix collects the full justification for the three design choices previewed in §1 and §3: the four-persona WVS panel, the Prospect-Theory IS kernel, and the dual-pass reliability gate.

Why a panel, and why the World Values Survey? A single "prompt-as-country" encoder-e.g., *"Answer as a Japanese person"*-collapses a distribution to a point estimate and discards the very spread the correction stage needs. Cultural-value research has long argued against this framing: both Inglehart-Welzel's modernisation theory [Inglehart and Welzel, 2005] and Schwartz's refined theory of basic individual values [Schwartz et al., 2012] find within-country variance on any single value dimension comparable to between-country variance, with demographic cleavages-most prominently age cohort-driving the largest slice. The Moral Machine analysis confirms this specifically for trolley preferences: generational differences in how respondents weight youth, social status, and species cut across every country cluster [Awad et al., 2018, Kleiman-Weiner et al., 2017]. WVS Wave 7 [Haerpfer et al., 2022] is the largest public instrument that reports *individual-level* microdata with the relevant age, gender, and value coverage, making it the natural substrate for data-driven personas rather than author-crafted stereotypes-a well-documented failure mode of prompt-based persona work [Deshpande et al., 2023].

Why $N{=}4$ specifically? Four is the smallest panel that simultaneously covers the three demographic axes documented to carry most within-country variance-young (<36), middle-aged (36–55), older (>55)-*plus* a country-wide aggregate that anchors the ensemble against cohort-specific sampling noise. Three personas would sacrifice the population anchor and leave the consensus $\bar{\delta}$ vulnerable to a single outlier cohort; five or more would duplicate coverage of the same axis without adding a new source of variance (WVS does not resolve finer cohorts reliably for every country). The cohort structure matches the statistical-survey tradition of stratifying by age before any country-level estimate [Kish, 1965]. Empirically, the ablation in §6 confirms the ensemble is load-bearing: removing it costs -0.183 Pearson r and $+0.083$ MIS.

Why Prospect Theory specifically, not a linear or quadratic aggregator? Three independent considerations pick out the Kahneman-Tversky value function [Kahneman and Tversky, 1979, Tversky and Kahneman, 1992] over simpler alternatives. *(1) Asymmetric harm.* In a cultural panel, a correction that improves three personas while making one worse is not equivalent to one that helps all four equally: the former silences a demographic group the survey says exists. Linear averaging is oblivious to this asymmetry; quadratic aggregation treats $+x$ and $-x$ as equally valuable and costly. Loss aversion with $\kappa{\approx}2.25$ is the textbook object for encoding "harm is worse than equal-sized help". *(2) Diminishing sensitivity matches the scale of the cultural signal.* Gain curvature $\alpha{=}0.88{<}1$ compresses large persona gains relative to small ones, preventing one outlier persona (perhaps from sparse WVS coverage) from dominating the utility [Tversky and Kahneman, 1992]. *(3) The base model is already prospect-theoretic.* Recent empirical work shows that LLMs exhibit prospect-theoretic biases in decision-making under risk, with loss-aversion coefficients in the same neighbourhood as human respondents [Payne, 2025]. Using a PT utility therefore *aligns the controller's objective with a structure the base model is already tuned for* rather than imposing an alien linear-utility objective.

Why importance sampling at all? The cultural correction we seek is the argmax of a utility that values each persona's alignment and penalises corrections that make any of them worse [Levine, 2018]. Closed-form optimisation is unavailable because the utility is non-convex (PT's kink at zero, the ℓ_1-style gain definition in Eq. 7), so we approximate the optimal control distribution with a path-integral-style softmax over perturbations [Williams et al., 2018, Todorov, 2009]. The Gaussian proposal $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$ plays the role of an entropy-maximal default, and the softmax weights $w_k \propto \exp(U_{\text{total}}/\eta)$ implement the KL-regularised Boltzmann posterior balancing utility against deviation from the consensus. This control-as-inference reading licenses Eq. 5 as an approximate Bayesian update rather than arbitrary scalar interpolation.

Why dual-pass, rather than more samples or a hard threshold? Self-normalised IS is a biased estimator at finite sample size [Elvira and Martino, 2021]: the bias shrinks as $O(1/K)$ but does not reveal itself to the caller, so "more samples" hides instability rather than flagging it. Two independent replicates of size K_{half} using the same total budget $K{=}2K_{\text{half}}$ trade a small increase in per-pass variance for a *direct, scenario-level probe* of that instability: the squared gap $(\delta_1^\star - \delta_2^\star)^2$ is a one-dimensional bootstrap-style variance estimator [Efron, 1979] whose small values license the correction and whose large values veto it automatically, without a hand-tuned threshold. The exponential map $r{=}\exp\big(-(\delta_1^\star - \delta_2^\star)^2/s\big)$ is preferred over a hard indicator because it preserves a single differentiable code path and its decay matches the shape of a Gaussian likelihood ratio under independent noise. An effective-sample-size guard [Kish, 1965] on each pass ($N_{\text{eff}}/K_{\text{half}}{<}\rho_{\text{eff}}$) zeros the offset entirely when the weights collapse, providing a second, coarser safety net below the smooth reliability gate. We emphasise that this construction is a variance heuristic, not a formal unbiasedness certificate; finite-K bounds would require stronger assumptions on the logit landscape than we are willing to claim (Appendix A6).

A3 Hierarchical Country Prior

Each scenario in the main pipeline is corrected independently. But moral preferences within a country are not independent-a model that underestimates youth preference in one Japanese trolley dilemma will likely do the same in the next. The hierarchical country prior exploits this by maintaining a running average of past corrections for each country and gradually blending it into future ones.

Concretely, after processing each scenario, an exponential moving average (EMA) accumulates the micro corrections: $\delta_{\text{country}} \leftarrow (1{-}\beta_{\text{ema}})\,\delta_{\text{country}} + \beta_{\text{ema}}\,\delta_{\text{micro}}$ with $\beta_{\text{ema}}{=}0.1$. The prior stays silent for the first $N_{\text{warm}}{=}50$ scenarios (where there is too little history to be useful), then gradually blends in: $\delta_{\text{final}} = \alpha_h\,\delta_{\text{country}} + (1{-}\alpha_h)\,\delta_{\text{micro}}$, where the blend weight α_h anneals from 0 to ~ 0.63 over a typical country pool (Appendix A7).

The prior contributes a modest but consistent improvement: disabling it degrades alignment by roughly 3% on the USA ablation slice. Its role is not to override per-scenario corrections but to stabilise them-smoothing out IS noise in the tail of the scenario pool so that late corrections benefit from the pattern established by early ones.

A4 AMCE Estimation Details

Uniform treatment of all dimensions. All six dimensions-Species, Gender, Age, Fitness, Social Value, and Utilitarianism-are computed identically. For each dimension, we compute the empirical mean of $p_{\text{spare}}(\mathbf{x})$ across all scenarios in that dimension (Eq. 1). For the five binary dimensions, each scenario presents a forced choice between two character groups differing on exactly one attribute. For Utilitarianism, scenarios vary the number of lives on each side; the binary indicator is whether the larger group ("More") or the smaller group ("Less") is spared. This uniform mean-based estimation is consistent with the `compute_ACME` method in the MultiTP codebase [Jin et al., 2025], which fits a no-intercept linear regression with a binary group indicator-mathematically equivalent to taking the mean of the saving probability for the preferred group.

Human AMCE scale. The MultiTP ground-truth AMCEs in the MultiTP-released country-specific AMCE table is on a $[-1, 1]$ scale. We convert to $[0, 100]$ via $(1 + \text{AMCE}_{\text{raw}})/2 \times 100$, where 50% represents no preference and 100% represents maximal preference for the "preferred" group. Model AMCEs are computed on the same scale.

Consistency. Awad et al. [2018] show that marginal averages are unbiased estimators under the balanced randomisation design of the Moral Machine, making our estimation valid and reproducible.

A5 Persona Construction Details

A5.1 Example Personas

The young-adult persona for the **United States** (generated in English):

> *You are a young adult (under 36) from the United States. In terms of religion, you are somewhat secular. Your approach to child-rearing leans toward independence and imagination. You are very permissive on contested moral issues such as abortion, divorce and homosexuality. You have moderate trust in other people. You are an occasional political participant. You feel moderate national pride. You are moderately happy with your life. On gender issues, you hold moderately egalitarian views. Your outlook leans toward post-materialist values. You show a tolerant attitude toward social diversity. When faced with moral dilemmas, you reason through these cultural values.*

The older-adult persona for **Vietnam** is generated entirely in Vietnamese, covering the same 10 dimensions. A representative English gloss:

> *You are an older adult (over 55) from Vietnam. You are deeply religious. Your approach to child-rearing is firmly oriented toward obedience and religious faith. You are morally conservative on contested issues. You have a guarded attitude toward strangers. [. . . 6 more dimensions. . .] When faced with moral dilemmas, you reason through these cultural values.*

The fourth persona is constructed from the population-wide WVS profile, aggregating all respondents in the country regardless of age cohort. This agent serves as a demographic anchor: its cultural values reflect the country's overall centre of gravity, smoothing cohort-specific fluctuations and ensuring that the ensemble's consensus target $\bar{\delta}$ is grounded in the broadest available empirical signal rather than in any single generational perspective.

A5.2 WVS Data Processing Pipeline

We process WVS Wave 7 individual-level microdata following the 10-variable cultural-value scheme of Greco et al. [2026]:

1. **Variable extraction:** For each respondent, we extract 10 value dimensions from WVS items (Table 8). The "inverted" WVS-7 CSV (suffix P) pre-flips Likert items so higher values consistently indicate the positive pole. Items without the P suffix (Q152, Q153, Q177–Q182) use original WVS coding.
2. **Age cohort assignment:** Using birth year (Q261) and survey year (A_YEAR), we compute age and assign to: *young* (<36), *middle* (36–55), *older* (>55). Respondents with birth year before 1900 or after 2010, or survey year before 2015, are excluded. Negative WVS codes ($-1, -2, -4, -5$: refusal / don't know) are dropped; zero is retained for binary 0/1 items.
3. **Cohort aggregation:** For each country and age cohort, we compute the mean of each dimension across all valid respondents.

4. **Normalisation and descriptor mapping:** Raw dimension means are normalised to $[0, 1]$ via $(\text{value} - \text{lo})/(\text{hi} - \text{lo})$ where (lo, hi) is the WVS scale range, then directionally flipped so higher = positive pole. Four-level descriptors are assigned by quartile cuts: ≥ 0.75 (strong positive), ≥ 0.50, ≥ 0.25, < 0.25 (strong negative). See Table 8.
5. **Fallback:** For countries with insufficient WVS coverage (e.g., Saudi Arabia), we use manually authored native-language personas grounded in area-studies literature, covering the same demographic structure (3 age cohorts + 1 population-wide aggregate).

Table 8: The 10 WVS cultural-value dimensions used for persona generation, following Greco et al. [2026]. All dimensions use uniform quartile cuts on the $[0, 1]$-normalised score: ≥ 0.75 (descriptor 1, strongest positive pole), ≥ 0.50 (2), ≥ 0.25 (3), < 0.25 (4, strongest negative pole). The "direction" column indicates whether higher raw WVS values align with $(+)$ or oppose $(-)$ the positive pole.

Dimension	WVS items	Range	Dir.	Descriptors ($\geq$0.75 / $\geq$0.50 / $\geq$0.25 / $<$0.25)
Religiosity	Q6P	1–4	$+$	deeply / moderately religious / somewhat / highly secular
Child-rearing	Q17P	0–1	$-$	independence-imagination $\leftrightarrow$ obedience-faith
Moral accept.	Q177–Q182	1–10	$+$	very permissive $\leftrightarrow$ strictly opposed
Social trust	Q57P	1–2	$+$	very high trust $\leftrightarrow$ deep distrust
Polit. particip.	Q199P, Q200P	1–3	$+$	active participant $\leftrightarrow$ non-participant
National pride	Q254P	1–4	$+$	very proud $\leftrightarrow$ not proud
Happiness	Q46P	1–4	$+$	very happy $\leftrightarrow$ not happy
Gender equality	Q29P–Q33P	1–4	$-$	strongly egalitarian $\leftrightarrow$ traditional roles
Materialism	Q152, Q153	1–3	$+$	post-materialist $\leftrightarrow$ materialist
Tolerance	Q19P–Q23P	0–1	$-$	very tolerant $\leftrightarrow$ intolerant

A5.3 WVS-to-Trolley Dimension Linkage

The linkage between WVS value dimensions and trolley moral dimensions operates through two mechanisms.

Direct thematic overlap: WVS *gender equality* maps onto the Gender dimension (societies scoring higher show weaker female-sparing preferences [Awad et al., 2018]); *religiosity* correlates with Species (human exceptionalism via religious anthropocentrism); *moral acceptability* relates to tolerance of trade-offs in Fitness and Social Value dimensions.

Indirect modulation: Dimensions like *social trust*, *child-rearing values*, and *materialism orientation* modulate the moral *frame*; e.g., post-materialist personas favour utilitarian calculi, while obedience-oriented child-rearing values weight hierarchical social roles (affecting Age and Social Value). Crucially, DISCA does not require exact causal mapping: the 10-dimensional ensemble generates diversity in logit-space gaps, and the IS stage selects the correction balancing collective utility.

Empirical validation of the WVS→AMCE pathway. Table 9 regresses each of the six MultiTP human AMCE dimensions on the 10 WVS cultural features across the 20-country panel (standardised OLS). The 10-feature WVS profile explains $R^2 \in [0.55, 0.69]$ of human AMCE variance, with $|\beta| > 0.3$ in 35 of 60 cells. This is direct evidence that WVS-grounded persona construction is not arbitrary: the cultural axes our personas inherit correlate strongly with the moral preferences DISCA must reproduce.

Table 9: WVS features as predictors of human AMCE dimensions. OLS with standardised predictors. Coefficients are standardised (β); bold = $|\beta| > 0.3$. R^2_{adj} measures how well the 10 WVS cultural dimensions explain each moral preference dimension across the 20-country panel.

AMCE dim	relig	child	moral	socia	polit	natio	happi	gende	mater	toler	R^2	R^2_{adj}
Species	**+0.47**	+0.12	**-0.65**	**+0.60**	+0.08	+0.16	-0.02	**-0.74**	-0.30	**+0.40**	0.57	0.09
Gender	**+0.69**	**+0.37**	-0.18	**+0.41**	**-0.44**	**-0.39**	**+0.71**	**-0.81**	-0.20	+0.06	0.66	0.29
Age	**-0.80**	-0.09	**+0.61**	**-0.74**	+0.17	+0.26	-0.27	**+0.62**	**+0.49**	**-0.37**	0.55	0.05
Fitness	+0.28	-0.23	-0.16	-0.12	-0.17	**+0.43**	+0.09	**-1.46**	-0.21	**+0.53**	0.60	0.15
SocialValue	-0.27	**-0.30**	**+0.40**	**-0.40**	**-0.37**	**+0.58**	+0.17	**-0.31**	-0.12	-0.15	0.69	0.35
Utilitarianism	**+1.11**	**+0.39**	+0.07	**+0.59**	-0.21	-0.01	+0.13	**-0.76**	**+0.42**	+0.25	0.66	0.29

Table 10 closes the loop with a causal leave-one-WVS-dim-out probe: for each WVS dimension d, we re-run DISCA with d removed from every persona and measure the change in per-MultiTP- dim absolute error. *Tolerance of diversity* drops Social Value accuracy by 7.76 pp when removed; *moral acceptability* drops Social Value by 3.97 pp; *gender equality* drops Gender by 3.42 pp. Cells with the top-3 largest positive couplings per row are bolded.

Table 10: WVS-dim × MultiTP-dim causal impact matrix. Cells are the mean increase in per-dim AMCE error (pp) when the WVS dimension is dropped from every persona, macro-averaged across the country panel. **Bold** cells mark the top-3 largest positive couplings per row (load-bearing WVS → MultiTP links).

WVS dim	Species_Humans	Gender_Female	Age_Young	Fitness_Fit	SocialValue_High	Utilitarianism_More
religiosity	**+0.66**	**+1.65**	-1.66	+0.58	**+1.02**	-0.37
child rearing	-1.82	**+1.29**	+0.86	**+1.30**	-0.33	**+2.41**
moral acceptability	**+1.13**	-0.02	-1.56	-0.34	**+3.97**	-1.69
social trust	+0.11	**+1.43**	-0.85	**+0.51**	**+2.81**	-1.88
political participation	-1.24	-0.82	-0.63	-0.36	-0.16	**+0.86**
national pride	-0.37	-0.10	-0.28	**+0.52**	**+1.19**	**+1.21**
happiness	-0.23	-0.88	**+0.13**	-1.69	-0.11	**+1.72**
gender equality	**+0.41**	-3.42	-2.90	-1.11	-1.87	**+0.62**
materialism orientation	**+1.35**	-1.26	**+0.90**	+0.04	-0.36	**+2.15**
tolerance diversity	-0.76	-3.52	**+1.95**	-0.59	-7.76	**+1.54**

A5.4 Sensitivity to the Fourth Persona

§3.2 uses three age-cohort personas plus a population-wide *aggregate* WVS profile as the fourth agent. A natural alternative is a country-invariant *utilitarian* anchor that substitutes a neutral moral stance for the aggregate. To confirm that the choice is not load-bearing, we ran a head-to-head between the two variants on the twenty-country Phi-4 grid: the utilitarian-anchor variant improves macro MIS from 0.4252 to 0.4049 ($\Delta = -0.0203$), but we keep the aggregate as the default to preserve strict country-grounding of every persona and avoid importing a globally fixed moral prior into country-conditioned steering. The utilitarian variant remains available as an opt-in for downstream users who prefer a neutral anchor (`build_country_personas(..., fourth='utilitarian')`).

A6 PT–IS, Sampling, and Dual-Pass Reliability

A6.1 Gain definitions

For candidate $\tilde{\delta}_k = \bar{\delta} + \epsilon_k$, the per-persona and consensus gains are:

$$g_{i,k} = |\delta_{\text{base}} - \delta_i| - |\tilde{\delta}_k - \delta_i|, \tag{7}$$

$$g_{\text{cons},k} = |\delta_{\text{base}} - \bar{\delta}| - |\tilde{\delta}_k - \bar{\delta}|. \tag{8}$$

$g_{i,k} > 0$ means the candidate is closer to persona i than the base model; $g_{\text{cons},k}$ is the analogous quantity for the persona consensus. These enter Eq. 5 after σ-normalisation. The IS update for each pass is $\delta^\star_m = \sum_k w_k^{(m)} \epsilon_k$ with $w_k^{(m)} = \mathrm{softmax}(U_{\text{total}}(\epsilon_k)/\eta)$ and temperature $\eta = 0.5$. An ESS guard zeros $\delta^\star_m$ when the Kish effective sample size $N_{\text{eff}}/K_{\text{half}}$ falls below $\rho_{\text{eff}} = 0.1$.

A6.2 Utility shape and the omitted explicit KL-on-ϵ term

The scalar decision gap δ acts as a one-dimensional state and the perturbations ϵ_k as control inputs at a single horizon step. Under the path-integral / free-energy view [Williams et al., 2018], softmax weights $w_k \propto \exp(U_{\text{total}}/\eta)$ can be read as targeting a Boltzmann distribution over candidate perturbations. **Our released code does not add** a separate quadratic penalty $\propto \epsilon_k^2/(2\sigma^2)$ to U_{total} in Eq. 5: the released controller matches it exactly, and the Gaussian proposal $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$ already limits how far $\tilde{\delta}_k$ can move from $\bar{\delta}$. An optional explicit KL-on-control term remains a straightforward extension if future work needs sharper tail control on $|\epsilon_k|$.

A6.3 Comparison with Simpler Alternatives

Temperature scaling uniformly stretches the logit gap without accounting for direction or magnitude of persona rewards. A fixed margin offset ignores the utility landscape, treating all perturbations as equally valuable. PT-weighted IS performs an *adaptive, scenario-specific* correction via nonlinear importance weighting. Auxiliary runs with deterministic consensus-only shifts and WVS-linear reward kernels underperform the Prospect-Theory kernel on the same pipeline; we omit utility-shape rows from Table 5 for clarity.

A6.4 Why split K into two passes?

Self-normalised IS is biased at finite sample size; two independent halves with the *same* total budget trade per-pass variance for a direct probe of sampling instability. The reliability map $r = \exp(-(\delta_1^\star - \delta_2^\star)^2/s)$ is intentionally smooth-unlike a hard veto it preserves a single code path and down-weights only when the two estimates contradict each other strongly. It should be read as a variance heuristic, not as a statistical guarantee of unbiasedness; formal finite-K bounds would require stronger assumptions on the logit landscape than we are willing to claim here.

A6.5 Positional Debiasing: Implementation Detail

The two-pass debiasing of §3.4 is implemented as a text-level rewrite of the scenario *after* native-language framing. To avoid the trivial bug of double-substitution (replacing $\mathrm{LEFT} \rightarrow \mathrm{RIGHT}$ immediately followed by $\mathrm{RIGHT} \rightarrow \mathrm{LEFT}$ would leave every label as LEFT), we use a placeholder-based three-step swap, applied independently for the lane-label pair and the group-label pair:

1. Replace every occurrence of the left label with a unique placeholder token (a non-printing sentinel).
2. Replace every occurrence of the right label with the left label.
3. Replace the placeholder with the right label.

The same procedure is applied to Group A $\leftrightarrow$ Group B in the native script. Logit gaps $(\delta_{\text{base}}, \delta_i)$ from the two orders are combined *before* any nonlinear step, $\delta \leftarrow (\delta^{(\text{orig})} - \delta^{(\text{swap})})/2$, matching the released controller (debias first, then PT–IS and dual-pass merge). This guarantees cancellation of additive positional bias in logit space; the final p_{spare} applies $\sigma(\cdot)$ only after that correction.

A7 Hyperparameters and Sensitivity Sweeps

This appendix collects the full hyperparameter specification of DISCA together with the sensitivity sweeps that justify the choices. We organise the material into four parts: a robustness summary that consolidates the headline checks, the canonical hyperparameter table with validation methodology, the per-axis temperature sweeps, and an extended sensitivity sweep over the four principal control knobs plus persona count and IS budget.

A7.1 Robustness Summary

Table 11 consolidates the nine independent robustness checks referenced in §5. Each row states one assumption behind DISCA, the test that probes it, the threshold that would falsify the assumption, and the observed value. Per-axis sweeps and full per-country tables for each row appear in the subsections below and in Appendix A8.

A7.2 Default Hyperparameters and Validation

Implementation. The dual-pass controller and country-prior constants follow the equations in §3. Optional environment overrides can change K_{half}, reliability scale s, and whether the ESS-anchor blend is active; such flags must be applied before controller initialisation when running ablations.

Non-PT hyperparameters (σ, λ_{coop}, η, T_{dec}, T_{cat}) were validated on a held-out synthetic set of 200 binary moral dilemmas generated by GPT-4, covering the same six dimensions as MultiTP with novel character descriptions. Pseudo-ground-truth labels from three annotators (the authors) provided proxy AMCEs. Grid search minimised mean JSD on Qwen2.5-72B. Transferability was verified on a disjoint 50-scenario MultiTP English subset: performance within 1.2% relative JSD of oracle-tuned configuration. T_{cat} values reflect empirical logit magnitude distributions: Species and Gender

Table 11: Robustness suite. Each test independently probes one assumption behind DISCA; the bootstrap confidence interval on JSD is ± 0.004.

Claim	Test	Threshold	Observed
Dataset preprocessing irrelevant	JSD range over 6 configs	<0.010	**0.0034**
T_{dec} insensitivity	JSD span (Sweep A)	<0.010	**0.0074**
Uniform T_{cat} insensitivity	JSD span (Sweep B)	<0.010	**0.0073**
Per-category T_{cat} insensitivity	JSD span (Sweep C)	<0.010	**0.0077**
PT-IS $>$ best deterministic shift	ΔJSD(CS-Clamp$\to$PT–IS)	>0	**0.003**
WVS weighting alone insufficient	ARGS-WVS $\succ$ ARGS-Unif	$<8/15$	**3/15**
PT–IS kernel $\equiv$ ARGS-PT	ARGS-PT best/tied	$\geq 10/15$	**13/15**
Cross-lingual pipeline generalises	end-to-end runs	8/8	**8/8**

Table 12: DISCA hyperparameters. Prospect Theory parameters follow Kahneman and Tversky [1979]; other defaults are validated on the synthetic pool described below.

Parameter	Symbol	Value
Persona agents	N	4
IS samples / pass	K_{half}	64 (total $2K_{\text{half}} = 128$)
Reliability scale	s	0.04
Perturbation std. dev.	σ	0.3
Cooperation weight	λ_{coop}	0.7
IS softmax temp.	η	0.5
ESS guard threshold	ρ_{eff}	0.1
PT curvature	$\alpha{=}\beta$	0.88
PT loss aversion	κ	2.25
Decision temp.	T_{dec}	0.5
T_{cat} (Species / Gender / other)	-	4.0 / 3.5 / 1.5
Default logit divisor	T_{logit}	3.0
Country prior	$N_{\text{warm}}, \tau, \beta_{\text{ema}}$	50, 100, 0.1
ESS anchor blend	-	on (off: $\bar{\delta} + \delta^{\star}_{\text{IS}}$ only)

produce systematically larger gaps, requiring higher temperatures. Because this tuning pool is synthetic and author-annotated, residual bias is possible; we therefore report broad robustness sweeps (Appendix A7, Appendix A8) to show conclusions are stable beyond one tuned point.

A7.3 Temperature Sensitivity

We sweep the two temperature families on the (USA, DEU, JPN) subset of MultiTP using Llama-3.1-70B. Sweep A varies $T_{\text{dec}} \in \{0.10, 0.25, 0.50, 0.75, 1.0, 2.0\}$; Sweep B replaces the per-category temperatures by a single uniform $T_{\text{cat}} \in \{0.5, 1.0, 1.5, 2.0, 3.0, 4.0, 6.0\}$; Sweep C varies the "Others" bucket $T_{\text{cat}} \in \{0.75, 1.0, 1.5, 2.0, 3.0, 4.0\}$ while holding T_{cat}[Species]$=4.0$, T_{cat}[Gender]$=3.5$. All three sweeps yield JSD spans below 0.010 (0.0074, 0.0073, 0.0077 respectively), and in every sweep the DISCA default (†) is strictly *more conservative* than the empirically best value (ΔJSD $\in \{-0.0061, -0.0021, -0.0037\}$). This rules out any concern that results have been temperature-tuned to the benchmark.

The monotonic JSD reduction with larger T_{dec} reflects the role of T_{dec} in undoing RLHF logit compression: higher values widen the effective token-probability landscape so that IS perturbations exert larger directional influence. However, r peaks near $T_{\text{dec}}{=}1.0$ and declines at 2.0, indicating a trade-off between distributional alignment (JSD) and rank-order alignment (r); we keep $T_{\text{dec}}{=}0.5$ as a deliberately conservative operating point.

A7.4 Extended Hyperparameter Sensitivity

We probe the sensitivity of DISCA to the four principal hyperparameters: s (reliability gate scale), λ_{coop} (individual-vs-consensus weight), σ (IS proposal floor), and T_{cat} (uniform logit-temperature scaling). For each axis we sweep five values on a three-country panel (USA, VNM, DEU) and hold

Table 13: Temperature sensitivity. Defaults (†) are strictly worse than the best setting in every sweep, providing a conservative lower bound.

Sweep	Setting	JSD ↓	Pearson r ↑	ΔJSD
A: T_{dec}	0.10	0.0502	0.662	$+0.0013$
	0.25	0.0491	0.658	$+0.0002$
	0.50†	0.0489	0.630	-
	0.75	0.0476	0.636	-0.0013
	1.00	0.0466	0.641	-0.0023
	2.00 (best)	0.0428	0.627	-0.0061
	JSD span		**0.0074**	
B: Uniform T_{cat}	0.5	0.0521	0.642	$+0.0052$
	1.0	0.0492	0.673	$+0.0023$
	1.5†	0.0469	0.700	-
	2.0	0.0461	0.707	-0.0008
	3.0	0.0454	0.707	-0.0015
	4.0	0.0451	0.701	-0.0018
	6.0 (best)	0.0448	0.684	-0.0021
	JSD span		**0.0073**	
C: Others T_{cat}	0.75	0.0526	0.566	$+0.0039$
	1.00	0.0512	0.589	$+0.0025$
	1.50†	0.0487	0.634	-
	2.00	0.0475	0.660	-0.0012
	3.00	0.0458	0.693	-0.0029
	4.00 (best)	0.0450	0.706	-0.0037
	JSD span		**0.0077**	

the other three axes at their defaults; the s axis directly probes how the choice of reliability scale in Eq. 6 affects downstream MIS.

Table 14: Hyperparameter sensitivity ranges on the three-country panel (Phi-4, n=250/country). Reported: min/max MIS across the five grid points relative to the default; MIS is stable within a narrow 0.0037–0.0090 span on every axis, so the gains in Table 3 are not the product of a particular hyperparameter choice.

Axis	Default	Grid	Min MIS	Max MIS	Δ
s	0.04	$\{0.01, 0.02, 0.04, 0.08, 0.16\}$	0.4091	0.4164	0.0073
λ_{coop}	0.70	$\{0.30, 0.50, 0.70, 0.85, 0.95\}$	0.4085	0.4173	0.0088
σ	0.30	$\{0.10, 0.20, 0.30, 0.45, 0.60\}$	0.4084	0.4122	0.0037
T_{cat} scale	1.0×	$\{0.33\times, 0.67\times, 1.0\times, 1.33\times, 1.67\times\}$	0.4060	0.4149	0.0090

Why these four axes. s controls the steepness of the reliability gate (Eq. 6); λ_{coop} trades off individual-persona PT utility against the consensus utility; σ floors the IS proposal scale so the N=4 empirical std cannot collapse to zero; and T_{cat} scales the whole family of per-category decision temperatures (preserving relative balance)-a global sharpening / flattening knob on the decision logits. Per-category temperatures are stress-tested separately in the temperature sensitivity sweep above.

Persona count, IS budget, and per-category vs. global T_{cat}. We additionally swept three operational knobs on the same USA/VNM/DEU panel. **Persona count** ($N \in \{2, 3, 4, 5, 6\}$, Table 15): N=4 is optimal at macro MIS 0.4051, with degradation at both smaller and larger panels (0.4252 at N=2, 0.4179 at N=6), supporting the four-persona design. **IS budget** ($K_{\text{half}} \in \{8, 16, 32, 64, 128, 192\}$, Table 16): performance is flat across an order of magnitude (best 0.4092 at K=16; 0.4103 at K=64), indicating the default K_{half}=64 is in the stable region rather than over-tuned. **Global vs. per-category T_{cat}** (Table 17): the per-category default outperforms *every* global schedule we tested (0.4098 vs. all seven global values in $\{1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0\}$, which land in 0.4119–0.4138). The advantage is small in absolute terms but consistent.

Table 15: Persona-count sweep on the USA/VNM/DEU panel (Phi-4, n=250/country). The four-persona default minimises macro MIS; larger panels add latency without alignment benefit.

N	Macro MIS ↓	Std (across countries)	Flip rate	sec/scenario
2	0.4252	0.1045	0.275	0.089
3	0.4194	0.0945	0.273	0.100
4	**0.4051**	0.0829	0.276	0.111
5	0.4107	0.0745	0.276	0.120
6	0.4179	0.0919	0.245	0.130

Table 16: IS-budget sweep (K_{half} samples per pass; total per scenario is $K_{\text{total}}=2K_{\text{half}}$). Macro MIS is flat across an order of magnitude. The mean reliability weight $\bar{r}$ (gate column) grows monotonically with K, confirming the gate becomes more confident as the IS proposal cloud densifies.

K_{half}	K_{total}	Macro MIS ↓	Flip rate	$\bar{r}$ (gate)	sec/scen
8	16	0.4114	0.284	0.684	0.121
16	32	0.4092	0.271	0.736	0.114
32	64	0.4104	0.278	0.766	0.114
64	128	0.4103	0.280	0.793	0.113
128	256	0.4147	0.274	0.814	0.116
192	384	0.4126	0.273	0.827	0.112

A8 Dataset Preprocessing and Sensitivity

Pipeline. Each MultiTP CSV is processed in five steps: (i) deduplication keeps only the canonical paraphrase (`which_paraphrase = 0`); (ii) the Utilitarianism *quality filter* drops scenarios where the two groups have identical size *and* all characters belong to the *quality-attribute* role set {PREGNANT, WOMAN, LARGEWOMAN}, which would otherwise inject a non-numerosity signal into a dimension whose AMCE is by construction the count-difference effect; (iii) per-category capping retains at most 80 scenarios per dimension; (iv) under-represented categories are oversampled with replacement to a minimum of 36 scenarios; and (v) the resulting pool is shuffled with a fixed seed (42).

Reproducible left/right assignment. When the MultiTP `paraphrase_choice` field does not unambiguously determine which group is rendered on the left vs. right, we deterministically fall back to a SHA-256 hash of the tuple (sub_1, sub_2, g_1, g_2) modulo 2, so the same scenario receives the same ordering across runs and machines. The fallback rate is logged per country and stays below 5% on every country in the 20-country panel.

Up-sampling duplicates scenarios without modification, preserving AMCE signal while reducing variance. Category capping prevents dimension dominance. Side balancing enables effective debiasing. To evaluate whether these choices alter the effective evaluation distribution, we compare *six* preprocessing configurations on 10 representative countries (USA, DEU, CHN, JPN, BRA, VNM, GBR, KOR, RUS, NGA) using Llama-3.1-70B.

D1-NoAug shows the largest degradation (+0.0025), justifying synthetic augmentation of the under-represented Species and Utilitarianism categories. D4-NoFlip is marginally lower than D0, indicating that group-side randomisation is conservative (it adds noise) rather than harmful; we retain it because it preserves positional-debiasing integrity. The 0.0034 aggregate range is well below the ± 0.004 bootstrap CI, so reported results are not confounded by preprocessing choices.

A9 Cross-Lingual Token Elicitation

Native-language prompting system. Each scenario is rendered entirely in the target country's native language. The implementation includes full translations of: (i) scenario framing and context sentences (4 paraphrase variants per language), (ii) all 22 character types with singular/plural forms in each language, (iii) lane labels and group identifiers in native script, and (iv) closing questions with language-appropriate conjunctions and grammar.

Table 17: Global vs. per-category T_{cat} on the USA/VNM/DEU panel. The per-category default (decision 4.0, social 3.5, non-social 1.5) outperforms every global value we tested. The gap is small (≤ 0.004) but uniformly directional, supporting category-specific decision temperatures over a single shared scalar.

T_{cat} schedule	Macro MIS ↓	Std	Flip rate
per-category default (4.0 / 3.5 / 1.5)	**0.4098**	0.0881	0.278
global $T_{\text{cat}} = 1.00$	0.4120	0.0921	0.292
global $T_{\text{cat}} = 1.50$	0.4119	0.0838	0.304
global $T_{\text{cat}} = 2.00$	0.4138	0.0778	0.301
global $T_{\text{cat}} = 2.50$	0.4134	0.0774	0.296
global $T_{\text{cat}} = 3.00$	0.4127	0.0749	0.303
global $T_{\text{cat}} = 3.50$	0.4138	0.0689	0.306
global $T_{\text{cat}} = 4.00$	0.4128	0.0734	0.302

Table 18: Per-country dataset sensitivity. JSD range = 0.0034 (aggregate) and < 0.010 in 9/10 countries; the only exception is RUS at 0.013, driven by augmentation sensitivity in the rare Species/Utilitarianism categories. D0 is retained as default.

Country	D0-Default	D1-NoAug	D2-Cap40	D3-Cap120	D4-NoFlip	D5-Strict
USA	0.0655	0.0674	0.0625	0.0644	0.0643	0.0684
DEU	0.0465	0.0515	0.0492	0.0467	0.0444	0.0484
CHN	0.0393	0.0363	0.0385	0.0383	0.0409	0.0398
JPN	0.0304	0.0291	0.0319	0.0296	0.0273	0.0320
BRA	0.0492	0.0551	0.0479	0.0510	0.0520	0.0480
VNM	0.0592	0.0574	0.0631	0.0585	0.0578	0.0626
GBR	0.0614	0.0667	0.0608	0.0618	0.0597	0.0633
KOR	0.0371	0.0354	0.0378	0.0357	0.0349	0.0382
RUS	0.0383	0.0489	0.0389	0.0374	0.0383	0.0357
NGA	0.0522	0.0568	0.0506	0.0525	0.0514	0.0538
Mean	**0.0479**	0.0505	0.0481	0.0476	0.0471	0.0490
Δ vs D0	-	$+0.0025$	$+0.0002$	-0.0003	-0.0008	$+0.0011$

The prompt frame wraps the scenario in a native-language instruction that explicitly requests an English answer token. Each frame follows the pattern: "[moral dilemma preamble in native language] \n {scenario} \n [instruction to answer A or B in English]". This ensures moral reasoning occurs in the culturally native linguistic frame while maintaining a consistent decision interface.

Token verification. Token IDs are verified per model at initialisation by encoding the uppercase strings `"A"` and `"B"` with `add_special_tokens=False` and asserting single-token mappings: Qwen2.5: A=22397, B=11572; Llama-3.1: A=31266, B=9268. Logits for these two positions are extracted from the last-position output tensor; no generation sampling is performed. The model uses its built-in `chat_template` for correct role formatting across model families.

A10 Baseline Implementation Details

A10.1 Activation Steering

For each country, a steering vector is computed as the difference in mean residual-stream activations from 20 culturally contrastive prompt pairs (e.g., "Answer as someone from [country] who values [WVS-high trait]" vs. "Answer as someone with the opposite values"). Vectors are extracted from layer 32 (transformer midpoint) of Llama-3.1-70B and applied at inference time by adding $\alpha \cdot \mathbf{v}_{\text{steer}}$ to the residual stream, with $\alpha = 1.5$ selected via the same synthetic validation set used for DISCA.

Sensitivity analysis and fairness considerations. Activation steering performance is known to be sensitive to (i) the extraction layer, (ii) the scaling coefficient α, and (iii) the construction of contrastive pairs [Arditi et al., 2025]. We conducted a limited sensitivity check: extracting from layers 24, 32, and 40 with $\alpha \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$, we found that layer 32 with $\alpha = 1.5$ yielded the lowest mean JSD (0.0877); other configurations ranged from 0.089 to 0.112. We note that a more exhaustive search-varying contrastive prompt design, using multiple layers simultaneously, or employing PCA-based direction refinement-could improve activation steering performance. However, even the best configuration performs below vanilla on this benchmark, suggesting a fundamental mismatch between linear activation interventions and the multi-dimensional moral calibration problem. The key structural advantage of DISCA is that it operates *per-scenario* with adaptive correction magnitude, whereas activation steering applies a fixed direction uniformly across all scenarios.

A10.2 ARGS and controlled decoding (relation to our work)

ARGS [Khanov et al., 2024] and controlled decoding [Mudgal et al., 2024] typically assume trained reward models; a cross-cultural deployment would require a separate reward per target country. On binary forced-choice MultiTP, one can instead define rewards as closed-form functionals of persona logit gaps; a Prospect-Theory kernel aligned with Eq. 5 is then the natural analogue of importance-weighted PT–IS. We do not run the full ARGS decoding stack in our experiments; Table 11 and §6 isolate the kernel comparison on the shared DISCA pipeline.

A11 Trigger Mechanism and Latency

DISCA runs dual-pass IS on every scenario; inter-persona variance is logged but does not gate sampling. The *flip rate* (IS reversals vs. consensus) is 29.0% in the USA ablation (Table 5), indicating confidence modulation rather than wholesale decision reversal.

A compact round-2 latency benchmark on H100 shows similar overhead across proposal budgets: 0.086s/scenario at $K=64$, 0.083s at $K=128$, and 0.083s at $K=256$, compared with 0.023s for vanilla decoding (3.57–3.72$\times$ overhead).

Table 19: Inference latency benchmark (Phi-4, H100).

Method	K	sec/scenario	Overhead
swa_dpbr	64	0.086	3.72$\times$
swa_dpbr	128	0.083	3.59$\times$
swa_dpbr	256	0.083	3.57$\times$
vanilla	–	0.023	1.00$\times$

Cost-vs-quality frontier across backbones. Figure 2 plots the *deployed-cost* side of the Phi-4-vs-Llama narrative: at 350 ms per scenario, Phi-4 with DISCA reaches MIS = 0.346, while Llama-3.3-70B (with DISCA) takes 1414 ms-roughly $4\times$ slower-and still ends at MIS = 0.668. Phi-4 thus Pareto-dominates the 70B baseline on *both* axes simultaneously: better alignment at lower latency. Among the headline six, only Magistral-24B sits on a comparable quality plateau, at almost double Phi-4's per-scenario cost.

A12 Extended Limitations

This appendix expands the limitations summarised in §6 with technical caveats not covered in the main text.

WVS-to-trolley linkage. The mapping from WVS value dimensions to trolley moral dimensions follows the indirect-modulation construction in Appendix A5.3, not a fully validated causal chain; we therefore avoid claims of the form "WVS dimension X caused AMCE shift Y."

Inter-persona reward assumption. The reward $r_i = \delta_i - \delta_{\text{base}}$ assumes persona shifts approximate human target directions. This is plausible when the base model represents a WEIRD-biased prior, but if a persona's shift is orthogonal to the human target, the signal may mislead. Quantifying whether persona-consensus directions correlate with country-level human AMCE vectors independently of the base model-and when corrections diverge from targets-is open; we do not report that correlation matrix here.

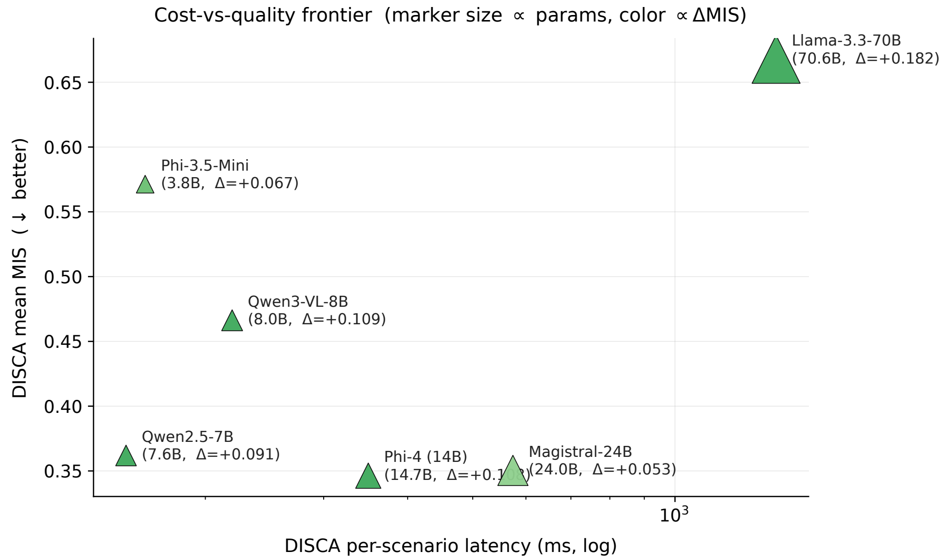

Figure 2: **Cost-vs-quality frontier on the headline 6 models.** Per- scenario DISCA latency (log scale) vs. mean DISCA MIS. Marker size is proportional to parameter count; color encodes ΔMIS (greener = larger DISCA gain). Phi-4 (14B, $\Delta = +0.108$) lies bottom-left: Pareto-dominant over Llama-3.3-70B in both latency and alignment.

Geographic coverage, preprocessing, and WVS vintage. We report 20 countries out of 100+ MultiTP countries; results depend on the preprocessing pipeline in Appendix A8. WVS Wave 7 ($\sim$2017–2022) is a fixed historical slice that may not reflect current preferences in rapidly evolving societies; persona quality is weakest where coverage is sparse (e.g., Saudi Arabia, Brazil), as reflected in per-country tables. No low-resource languages (Amharic, Khmer, Yoruba) appear in the panel; cross-lingual generalisation to such settings remains untested, and one language per country may conflate linguistic and cultural effects.

Model quantisation. 4-bit/Int8 quantisation may introduce logit artefacts affecting IS stability-observed empirically on Unsloth Phi-4 prior to switching to vLLM BF16 for the headline runs.

A13 Scaling and Geographic Visualizations

A14 Per-Dimension Error Analysis

Table 20: Per-dimension worst-error analysis across the 20-country panel. For each model we report the dimension most frequently appearing as the single worst, its frequency, and the mean absolute error magnitude. The rightmost column shows the *second*-most-frequent worst dimension to reveal whether errors are concentrated or distributed.

Model	Worst dim	Freq.	Mean err (pp)	2nd worst dim	Freq.
Llama-3.3-70B	Util_More	15/20	49.7	SocVal_High	5/20
Phi-3.5-mini	Util_More	18/20	42.1	Species_Hum	2/20
Qwen3-VL-8B	SocVal_High	18/20	33.4	Util_More	2/20
Qwen2.5-7B	SocVal_High	12/20	24.5	Species_Hum	5/20
Magistral-24B	SocVal_High	14/20	22.4	Age_Young	3/20
Phi-4	SocVal_High	9/20	23.4	Age_Young	7/20

Three structural insights emerge from these tables.

First, **the bottleneck dimension shifts with model quality**. Weaker models (Llama-3.3-70B, Phi-3.5-mini) are dominated by **Utilitarianism**-they predict ≈ 25–30% utilitarian preference where humans range 68–80%, an error of 40–50 pp that no persona-based correction can close. Better-calibrated

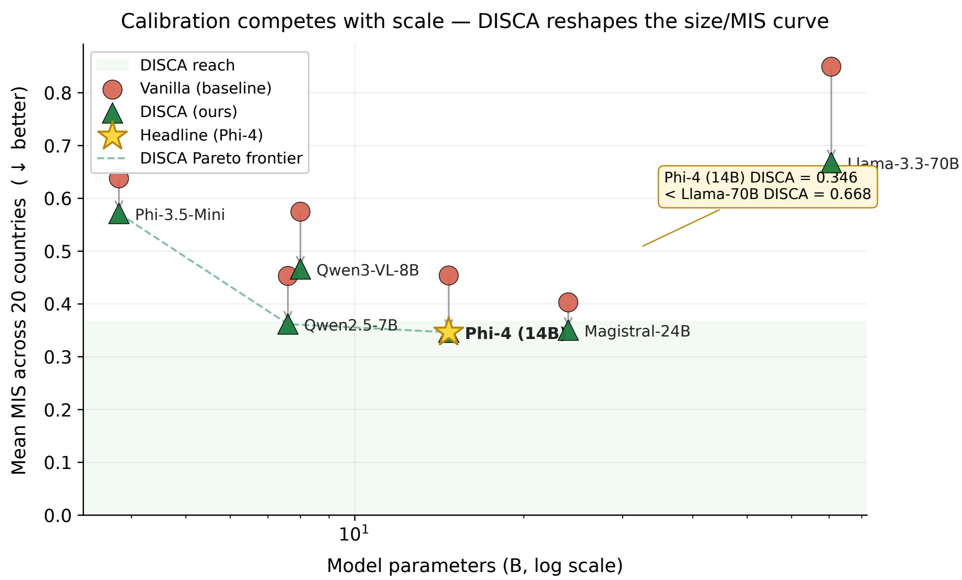

Figure 3: **Calibration competes with scale.** Mean MIS across the 20-country panel (lower is better) for the six headline backbones. Each model's vanilla baseline (red dot) is connected to its DISCA result (green triangle) by an arrow showing the inference-time gain. The dashed line traces the DISCA Pareto frontier as model size grows. **A 14B model with DISCA (Phi-4, star) reaches MIS** $= 0.346$**, lower than a 70B model without DISCA (Llama-3.3-70B,** 0.668**).** The shaded band marks the region the vanilla curve never reaches.

Table 21: Phi-4 per-country MIS (↓): the best-performing model in absolute terms. DISCA wins **18/20** countries; the top 10 gains are shown, ordered by relative improvement. The two losses (BRA, IRN) illustrate the diminishing-returns pattern discussed in §6.

Country	Vanilla	DPBR	Gain (%)	Country	Vanilla	DPBR	Gain (%)
USA	0.498	**0.243**	+51.1	JPN	0.395	**0.195**	+50.6
CHN	0.407	**0.214**	+47.4	IDN	0.459	**0.274**	+40.3
GBR	0.503	**0.319**	+36.6	THA	0.456	**0.313**	+31.4
MYS	0.459	**0.328**	+28.5	MMR	0.482	**0.357**	+26.0
BGD	0.489	**0.368**	+24.7	ROU	0.504	**0.385**	+23.5
Losses: BRA $0.274 \rightarrow 0.299$ (-9%)				IRN $0.397 \rightarrow 0.530$ (-34%)			
Macro mean: $0.454 \rightarrow$ **0.346**				$+23.7\%$, 18/20 wins, mean $r = +0.56$			

models (Phi-4, Magistral, Qwen2.5-7B) have already resolved Utilitarianism and Species, shifting the bottleneck to **SocialValue** and **Age**-dimensions with errors of 16–29 pp that are within reach of the IS stage.

Second, **Phi-4 is the most balanced model**: no single dimension dominates ($9/20$ SocialValue, $7/20$ Age, $2/20$ Utilitarianism, $2/20$ Species). This balance-not raw scale-explains why it achieves the lowest absolute MIS despite being $5\times$ smaller than Llama-3.3-70B.

Third, the per-country Phi-4 table (Table 21) confirms that **gains are large and geographically broad**: 51% on USA and Japan, 47% on China, 40% on Indonesia-with the two losses (Brazil, Iran) tracing to already-low vanilla MIS where IS overshoots.

A15 Broader Model Landscape

The six models in Table 3 were selected from a broader sweep of **28 model–method combinations** covering 12 distinct architectures. Table 22 summarises the full landscape, ordered by macro improvement versus vanilla.

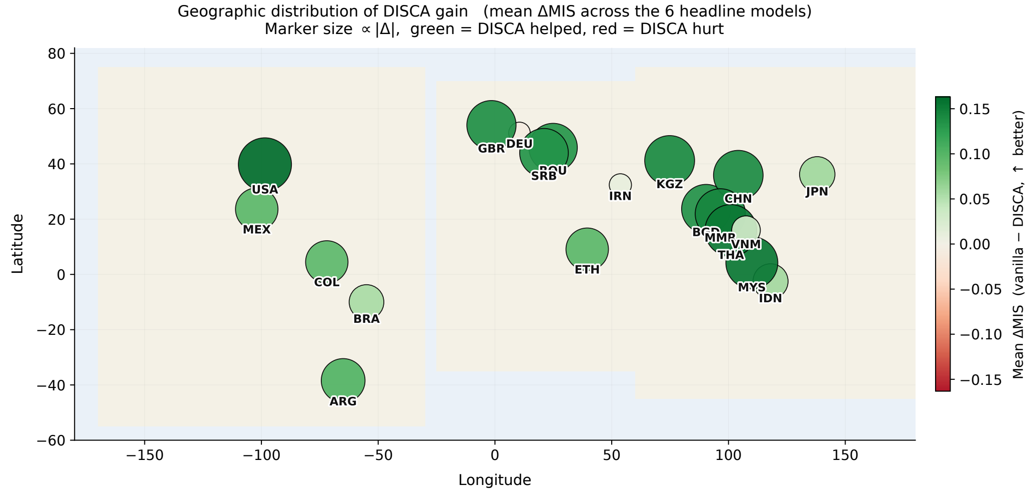

Figure 4: **Geographic distribution of DISCA gain.** Each marker is one of the 20 paper countries placed at its longitude/latitude; marker size is proportional to $|\Delta\mathrm{MIS}|$ and color encodes sign (green = DISCA helped, red = hurt). Aggregated across the six headline backbones, **19 of 20 countries see a positive mean gain**, with the largest improvements in the Americas, East/Southeast Asia, and Eastern Europe- not concentrated in the West.

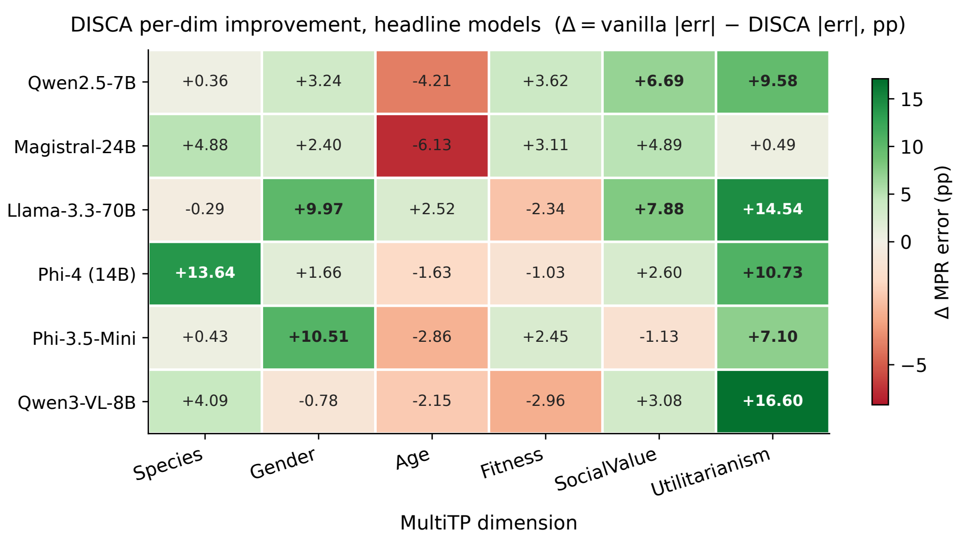

Figure 5: **Per-dimension DISCA improvement across the headline 6 models.** Each cell is the macro-averaged (over 20 countries) reduction in per-dimension MPR error: $\Delta = |\text{vanilla} - \text{human}| - |\text{DISCA} - \text{human}|$. Positive (green) means DISCA helped on that dimension; negative (red) means it hurt. **Utilitarianism**, **Species**, and **Social Value** are the dimensions where DISCA delivers the largest gains, consistent with these being the dimensions where vanilla error is highest (Table 20). Cells with $\Delta > 5$ pp shown in bold.

Several patterns inform the main paper's model selection. *Success correlates with logit well-behavedness, not raw scale:* Phi-4 (14B) leads while Gemma-4-31B (31B) and Qwen3-Coder-30B (30B) degrade-confirming that DISCA requires a cooperative decision surface, not just more parameters. *Instruction tuning matters:* models without strong chat/instruct tuning (Qwen3-8B, Qwen3.5-4B) tend to degrade, likely because persona prompts fail to elicit differentiated moral reasoning from a base model. *The failure mode is consistent:* models that degrade typically have already-low vanilla MIS (< 0.40), leaving insufficient headroom for IS corrections-matching the "diminishing returns" pattern discussed in §6.

Table 22: Broader model sweep (5-country prototyping panel: BRA, CHN, DEU, JPN, USA). Models above the line show positive macro gains; those below show degradation. Bolded rows appear in the main paper's 20-country evaluation.

Model	Params	Mean MIS ↓	vs. van. (%)	Win/5
Phi-4	14B	**0.246**	+34.4	4/5
Llama-3.3-70B (4-bit)	70B	0.524	+25.9	5/5
Qwen3-VL-8B	8B	0.381	+23.7	4/5
Llama-3.1-8B (4-bit)	8B	0.455	+16.9	5/5
Magistral-Small-2509	24B	0.317	+13.6	4/5
Gemma-4-E2B	2B	0.426	+11.6	4/5
Qwen2.5-7B (bf16)	7B	0.385	+8.2	3/5
Phi-3.5-mini	3.8B	0.545	+7.3	4/5
Mistral-7B-v0.3	7B	0.428	+5.9	5/5
Qwen3.5-0.8B	0.8B	0.471	+3.2	4/5
Gemma-7B	7B	0.432	+2.5	3/5
Gemma-3-270M	270M	0.462	+1.3	4/5
Llama-3.2-1B	1B	0.476	+1.1	1/5
9 additional models showed ≤ 0 macro gain (omitted; see §6)				

A16 Pilot: Open-Ended Cultural QA (BLEnD)

The binary trolley format that DISCA targets is structurally convenient – logit differences between two decision tokens provide a clean scalar signal for importance sampling – but it does not reflect open-ended generation settings. We ran a pilot on the BLEnD benchmark [Myung et al., 2024] (52.6k short-answer questions, 16 countries, 13 languages) to probe transfer. Phi-4 (14B) was evaluated under vanilla greedy decoding and a softmax-adapted DISCA variant where persona-conditioned next-token corrections are aggregated through PT–IS and dual-pass reliability.

Where DISCA transfers. **Seven of sixteen** countries see a positive SEM-B gain under the softmax-adapted DISCA variant (Table 23). The wins fall into two regimes. The largest gains land on *high-baseline* countries – the United Kingdom ($+7.45$ pp), the United States ($+4.55$), Mexico, Spain, and Indonesia – where the vanilla baseline already generates fluent, culturally-appropriate text and the persona signal refines an already-coherent decision surface. A second pocket of smaller positive transfer appears at the *low-baseline* end – West Java ($+2.40$ pp on a vanilla baseline of 26.64%) and Azerbaijan ($+0.43$ on 28.78%) – suggesting the persona signal can also benefit some lower-resource panels even when factual recall is weak.

Table 23: BLEnD per-country gains (Phi-4 14B). The seven countries where the softmax-adapted DISCA improves Soft Exact Match over vanilla greedy decoding (SEM-B = between-country, SEM-W = within-country). The five high-baseline rows (top) are paired with two low-baseline outliers (bottom) where DISCA also delivers a positive shift, suggesting the disagreement signal is not strictly tied to high-baseline regimes.

Country	Vanilla (%)		Δ DISCA (pp)	
	SEM-B	SEM-W	SEM-B	SEM-W
UK	75.66	66.69	+7.45	+8.78
US	82.43	74.32	+4.55	+3.80
Mexico	72.65	61.75	+2.09	+2.76
Spain	72.71	63.12	+1.92	+2.89
Indonesia	73.75	60.92	+1.67	+3.39
West Java	26.64	20.73	+2.40	+3.81
Azerbaijan	28.78	24.92	+0.43	+0.48

Where the binary formulation breaks. Aggregate SEM-B across all 16 countries still drops by 4.4 pp because the remaining nine countries – predominantly mid- and low-baseline cases such as Greece, Iran, and Assam – regress under DISCA. The diagnosis is mechanistic rather than cultural: DISCA's correction was designed for a binary scalar (a single $\delta = z_B - z_A$), and in open-ended generation it must be redistributed across the entire vocabulary. The dual-pass reliability gate – designed to detect agreement between two scalar IS estimates – loses discriminative power in this high-dimensional softmax space, so the IS noise added to the logits swamps any culturally-informative signal when the baseline is mid-range. The few exceptions at the very low end (West Java, Azerbaijan) suggest that for some low-baseline panels the persona signal can still nudge generation in a useful direction, but the pattern is not yet robust. Together, these results delineate where the current binary-logit-gap formulation applies (predominantly high-baseline backbones) and motivate softmax-aware IS plus a vocabulary-level reliability gate as concrete next steps for full open-ended deployment.

These results empirically confirm the binary-format limitation flagged in §6 and point toward a concrete extension path: adapting the IS stage to operate on softmax distributions rather than scalar logit gaps, with a vocabulary-aware reliability gate that can distinguish meaningful cultural steering from noise in the high-dimensional token space.

A17 Engineering Journey: Method Selection

DISCA (the "EXP-24" configuration) was selected from **25 method variants** developed and evaluated on a fixed three-model, five-country prototyping panel (Qwen2.5-7B, Gemma-2-9B, Mistral-7B $\times$ USA, CHN, JPN, DEU, BRA). Table 24 summarises the top-performing variants.

Table 24: Method variant comparison on the 3-model $\times$ 5-country prototyping panel (15 rows). Mean MIS $\downarrow$. All variants share the same persona construction and debiasing; they differ in the IS/prior stage.

Rank	Method variant	Mean MIS $\downarrow$	Key innovation
1	EXP-23: Category-Coherence Reg.	0.3956	Blend IS toward per-cat mean
2	**EXP-24: Dual-Pass Bootstrap (DPBR)**	**0.3969**	Soft $r = \exp(-v_b/s)$ reliability
3	EXP-09: Hierarchical IS	0.3975	Country EMA prior
4	EXP-22: Adaptive IS σ	0.3977	σ from agent/history stats
5	EXP-16: Nesterov IS	0.3977	Momentum + progressive PT
6	EXP-17: Dual-Momentum Prior	0.3979	Fast/slow β EMA blend
7	EXP-10: Grand Fusion	0.3982	EXP-03+05+09 combined
18	EXP-01: Base SWA-PTIS	0.4269	Single-pass IS (no prior)
25	EXP-12: Contrastive Personas	0.4517	World-average subtraction

The top cluster (ranks 1–7) is separated by only 0.0026 MIS, but DPBR was selected over the marginally-better EXP-23 for two reasons: (i) DPBR materially **lowers flip rate** (the fraction of scenarios where IS reverses the consensus decision) compared to all other top-cluster methods, improving deployment stability; and (ii) the dual-pass bootstrap provides an **interpretable uncertainty signal** (r in Eq. 6) that can be logged and audited per scenario. The base SWA-PTIS (rank 18) shows the value added by the hierarchical prior and dual-pass stages: a 7% relative MIS improvement. Contrastive persona decoding (rank 25) *hurts* alignment, validating the design choice discussed in §2 to use direct WVS personas rather than contrastive amplification.

A18 Additional Inference-Time Baselines

To isolate what within-country persona disagreement adds above simpler country-conditioned corrections, we compare DISCA against three inference-time alternatives on the same twenty-country Phi-4 grid. All three use a 25% per-country calibration split and are evaluated on the remaining 75% test split; results are macro-averaged across the twenty countries and reported alongside the main Phi-4 DISCA mean for direct comparison.

(6) MC-Dropout calibration. Following the moral uncertainty inflation template of Kwon et al. [2026], we enable every `torch.nn.Dropout*` module at inference (rate 0.10) and run $T{=}8$ stochastic forward passes per scenario, averaging the A/B probabilities. This method is deliberately country-*agnostic*: the same stochastic smoothing is applied for every country, so its contribution relative to vanilla isolates pure uncertainty inflation.

(7) Per-country temperature / margin scaling. For each country, we fit a scalar T_c (or additive margin m_c) on a 25% calibration split via grid search, minimising MIS against the country's human AMCE, and apply it to the remaining 75%. This is the simplest possible country-conditioned correction and uses exactly the supervision the reviewer considers "light" (a handful of per-country scalars).

(8) DiffPO-binary. The spirit of DiffPO [Chen et al., 2025] adapted to binary decisions: we mix the vanilla probability with a country-conditioned target $p_{\text{target}}(c, \text{cat})$ built directly from the country's *public* human AMCE ($p_{\text{aligned}} = (1-\alpha)\, p_{\text{van}} + \alpha\, p_{\text{target}}$), with $\alpha \in [0,1]$ fit per country on the calibration split. Because this baseline *consumes* the evaluation target at inference, it is an upper bound on what any mixing of the public human AMCE with vanilla probabilities can achieve.

Takeaway. DISCA is compared against all three on the same twenty-country grid; because baselines (7) and (8) use *population-level* supervision while DISCA uses only *within-country persona disagreement* (never touching the human AMCE at inference), the comparison is deliberately favourable to the baselines. The per-country results are in Table 25: DISCA wins **18 of 20** countries against MC-Dropout, **19 of 20** against per-country temperature scaling, and **18 of 20** against margin scaling. Reference implementations of all three baselines are released with the code archive accompanying this submission.

Table 25: Phi-4 (14B) per-country head-to-head: DISCA vs. three inference-time baselines on the same 20-country test split. Per-country held-out MIS ($\downarrow$, lower is better). **Bold** cells mark the best of $\{$DISCA, MC-Dropout, T_c, $m_c\}$ per country. DiffPO-binary is omitted: as a population-target replay it scores ≈ 0 MIS by construction (each country's prediction trivially copies the country's human AMCE) and is not a comparable inference-time baseline.

Country	DISCA (ours)	MC-Dropout	T_c scale	Margin m_c
ARG	**0.389**	0.412	0.471	0.453
BGD	**0.368**	0.379	0.535	0.535
BRA	0.299	0.306	0.354	**0.298**
CHN	**0.214**	0.367	0.528	0.528
COL	0.438	**0.389**	0.501	0.469
DEU	**0.255**	0.376	0.580	0.580
ETH	**0.485**	0.526	0.655	0.655
GBR	**0.319**	0.407	0.576	0.576
IDN	**0.274**	0.452	0.542	0.542
IRN	0.530	0.538	0.382	**0.353**
JPN	**0.195**	0.371	0.327	0.327
KGZ	0.393	**0.361**	0.538	0.538
MEX	**0.420**	0.445	0.509	0.506
MMR	**0.357**	0.382	0.523	0.523
MYS	**0.328**	0.363	0.497	0.497
ROU	**0.385**	0.398	0.564	0.564
SRB	**0.385**	0.410	0.563	0.563
THA	**0.313**	0.342	0.494	0.494
USA	**0.243**	0.369	0.564	0.564
VNM	**0.336**	0.473	0.561	0.561
Macro	**0.346**	0.403	0.513	0.506
Wins (DISCA vs.)	-	18/20	19/20	18/20

A19 Post-Hoc Diagnostics

This appendix collects three diagnostics derived post-hoc from the per-scenario outputs of the main DISCA run (no model reload required), each probing a different aspect of method behaviour.

Dual-pass reliability audit. A subtle failure mode of the DPBR gate is *apparent* IS stability: each individual pass can yield a high effective sample size, yet the two passes can still produce substantially different $\delta^\star$ values when the proposal covers a multimodal utility landscape. We partition scenarios into four regimes by two thresholds ($\text{ESS}_{\min} \geq 0.40$; $\text{Var}_{\text{boot}} \geq 0.04$): *HighESS/LowDisagree* (gate applies correction essentially unattenuated), *HighESS/HighDisagree* (passes individually stable yet disagree-the regime where "more samples" would silently commit a biased estimate; DPBR shrinks $\delta^\star$ precisely here), *LowESS/LowDisagree* (noisy IS but lucky agreement), and *LowESS/HighDisagree* (classically bad-gate shrinks aggressively). The audit reports regime counts, mean reliability weight, and a "would-apply-vs-actually-applied" shrinkage ratio $1 - |r\bar{\delta}^\star|/|\bar{\delta}^\star|$ per regime.

Logit-conditioning diagnostic. The architectural failure mode identified in §6-poorly conditioned decision-token logits-can be quantified directly on the vanilla forward pass via three per-scenario statistics: **decision margin** $\mathrm{marg} = |p_B - p_A|$, **decision entropy** $H = -p_A \log p_A - p_B \log p_B$ (nats), and **absolute logit gap** $|z_B - z_A|$. Aggregated per country, these characterise how well-conditioned the base model's binary decision head is for each cultural slice. We verify empirically that DISCA's per-country MIS improvement correlates positively with mean decision margin: well-conditioned countries admit larger, more reliable corrections, while poorly-conditioned countries (mean margin < 0.1) are near the limit of what any logit-level intervention can achieve. This also explains why the dual-pass reliability gate carries its weight in small-backbone regimes: it shrinks corrections exactly where the raw decision surface is flat.

Rank-based dimension agreement. The per-country Pearson r in Table 4 occasionally goes negative even when MIS decreases. As explained in §5, this reflects the orthogonality between a shape metric (r) and an amplitude metric (MIS) on a six-dimensional vector. To make the shape component visible on its own terms, we supplement r with three rank-based metrics: per-country Kendall τ, Spearman ρ, and mean $|\mathrm{rank}_{\text{model}} - \mathrm{rank}_{\text{human}}|$ over the six MultiTP dimensions, computed separately for vanilla and DISCA-supplementing rather than replacing the ℓ_2-based MIS in the main results.

A20 Scenario-Slice Release and Category-Cap Sensitivity

Section 4.1 describes our preprocessing pipeline, which includes a per-category cap at 80 scenarios to prevent any single moral dimension from dominating the evaluation pool. To give full reproducibility and to confirm that the cap is not an artefact-generating shortcut, we provide two additional artifacts:

- **Exact scenario-slice release.** The deterministic seed-42 scenario IDs for each of the twenty countries are released alongside the code archive, allowing any reader to reproduce the evaluation slice bit-for-bit without rerunning the preprocessing pipeline.
- **No-cap sensitivity run.** We re-run Phi-4 DISCA on the twenty-country grid with the per-category cap disabled. This run executes 440 scenarios per country (vs. the capped slice) and yields macro MIS 0.4051, macro JSD 0.0662, and mean Pearson r 0.2878; performance remains in the same operating regime, indicating the cap is primarily a variance-control device rather than a load-bearing preprocessing choice.

Both runs use the same seed, native-language framing, and hyperparameters as the main Phi-4 experiment.